HEILONGJIANG WANGYUNFENG WIND POWER PROJECT

Document Prepared By Longyuan (Beijing) Carbon Asset Management Technology Co.,Ltd.

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CLEAN DEVELOPMENT MECHANISM
PROJECT DESIGN DOCUMENT FORM (CDM-PDD)
Version 03 - in effect as of: 28 July 2006

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SECTION A. General description of project activity

A.1. Title of the project activity:

Heilongjiang Wangyunfeng Wind Power Project

Version: 02

Date: 01/08/2011

A.2. Description of the project activity:

Heilongjiang Wangyunfeng Wind Power Project (hereinafter refers to the proposed project) is developed by Hegang Longyuan Wind Power Co., Ltd. and located on Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province， P.R.China. The project has been started from Jun. 4th, 2009 and the preparation work of the project was conducted at present. It had been operation in the end of December, 2009.

The purpose of the project activity:

- ✧ The scenario existing prior to the start of the implementation of the Project is to provide the same annual electricity output as the Project by Northeast China Power Grid (NECPG).
- ✧ The Project is a newly-build wind plant with total installed capacity of 49.3MW (58*850kW). The project will generate electricity from wind resources using advanced wind power generation technology on a commercial basis and to deliver the electricity to the Northeast China Power Grid (NECPG). The implementation of the proposed project will achieve CO_2 emission reduction by replacing electricity generated by fossil fuel fired power plant.
- ✧ The baseline scenario of the Project is the same as the scenario existing prior to the start of implementation of the Project activity.

How the Project reduces GHG emissions:

The Project will utilize the wind resources of the local to generate electricity, which will be delivered to NECPG without CO_2 emissions. It is estimated that the feed-in electricity to the NECPG is approximately 120,069MWh per year with 2435.47h designed annual operation hours[1]. The Project activity will achieve greenhouse gas (GHG) emission reductions by avoiding CO_2 emissions from the business-as-usual scenario electricity generated by those fossil fuel-fired power plants connected into NECPG. It is estimated that annual emission reductions are 123,430tCO_2e per year.

Contributions of the project to sustainable development:

The Project will not only supply renewable electricity to grid, but also contribute to sustainable development of the local community and the host country by means of:

- ➢ To contribute to local economy development by providing electricity to meet local increasing energy demands;
- ➢ To reduce GHG emissions and to mitigate the emissions of other pollutants caused from local coal-fired power plants compared with a business-as-usual scenario by displacing part

[1] FSR of the proposed project

of electricity from fossil fuel-fired power plants;

- To be in accordance with the development priority of China energy industry, and to help diversify energy mix of NECPG by increasing the share of renewable energy;
- To create plenty of short-term employment opportunities and permanent jobs for the local people during the construction and operation period of the Project.

A.3. Project participants:

Name of Party involved (*) ((host) indicates a host Party)	Private and/or public entity(ies) project participants (*) (as applicable)	Kindly indicate if the Party involved wishes to be considered as project participant (Yes/No)
P.R.China (Host)	Hegang Longyuan Wind Power Co., Ltd. (the project owner)	No
United Kingdom of Great Britain and Northern Ireland	Total Gas & Power Ltd (the purchasing party)	No
(*) In accordance with the CDM modalities and procedures, at the time of making the CDM-PDD public at the stage of validation, a Party involved may or may not have provided its approval. At the time of requesting registration, the approval by the Party(ies) involved is required.		

Please see Annex 1 for detailed contact information

A.4. Technical description of the project activity:

A.4.1. Location of the project activity:

A.4.1.1. Host Party(ies):

People's Republic of China

A.4.1.2. Region/State/Province etc.:

Heilongjiang Province

A.4.1.3. City/Town/Community etc:

Damahe Forest Farm, Luobei County, Hegang City

A.4.1.4. Detail of physical location, including information allowing the unique identification of this project activity (maximum one page):

The proposed project is located in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, and P.R.China. It is 40 kilometres away from Luobei County, and 75 kilometres away from Hegang City. The central geographical co-ordinates of the project are: East longitude 130°39′56″ (E130.66556°) and north latitude 47°54′34″ (N47.90944°), and altitude 240m~430m.

Figure 1 shows the location of Heilongjiang Province and Luobei County. Figure 2 shows the location of the Project.

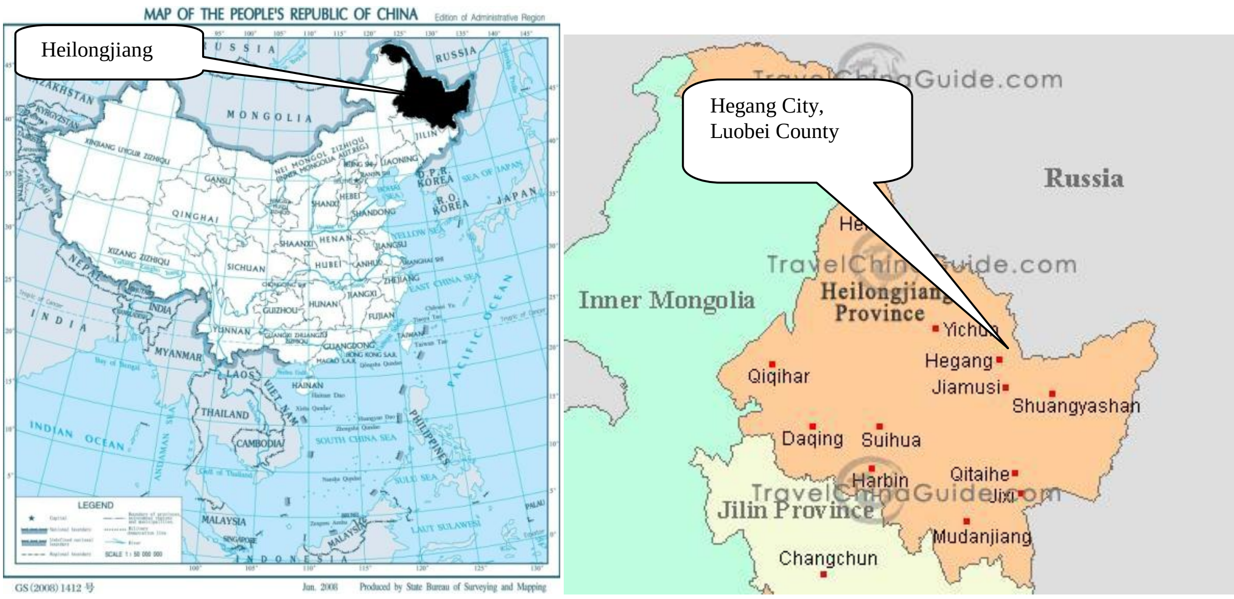

Figure 1 location of Heilongjiang Province and Luobei County

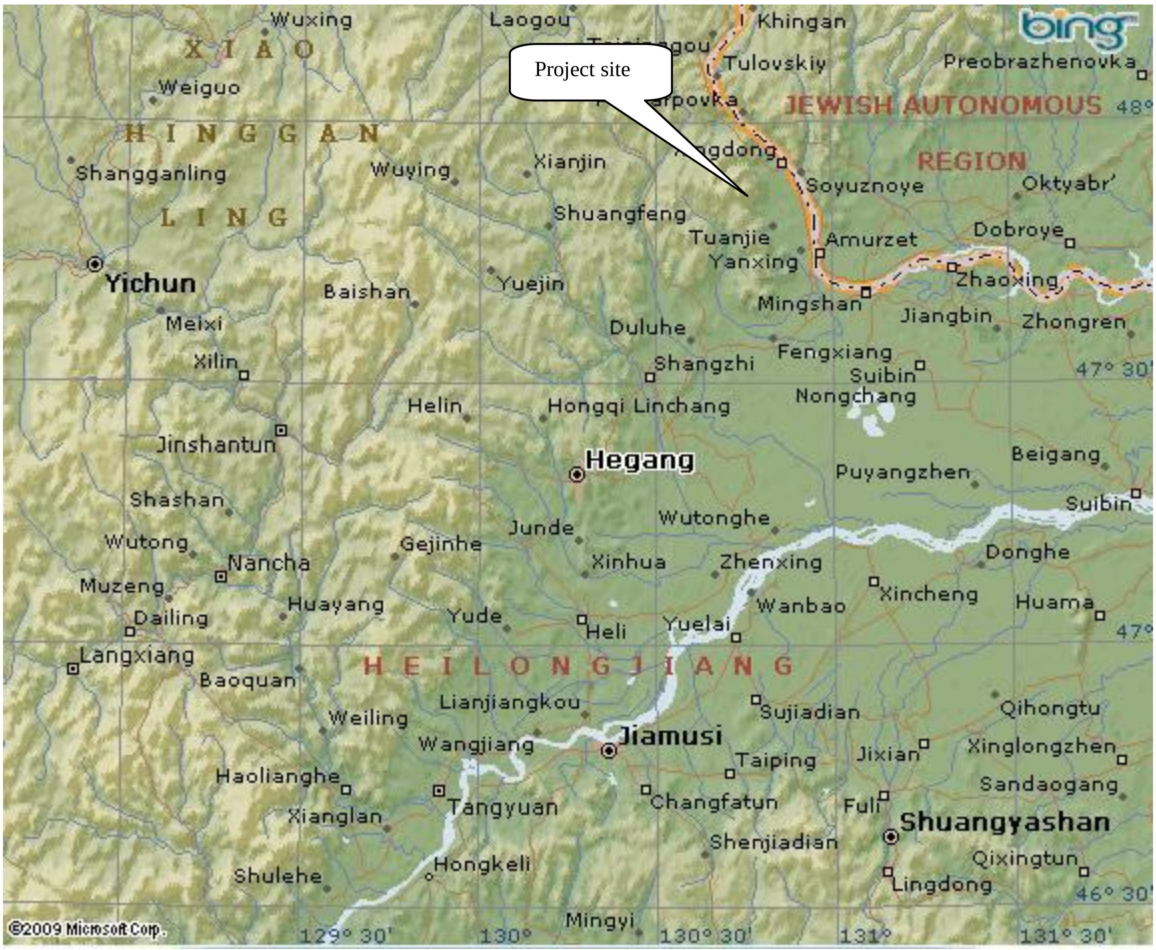

Figure 2 location of the Project

A.4.2. Category (ies) of project activity:

Category: Renewable Energy in grid connected applications

Sectoral Scope 1: Energy industries (renewable - / non-renewable sources)

A.4.3. Technology to be employed by the project activity:

(a) Technology to be employed before the Project:

The scenario existing prior to the start of the implementation of the Project is to provide the same annual electricity output as the Project by Northeast China Power Grid (NECPG); the key information about NECPG is listed as follows:

Emission Factor	$EF_{OM,y}$	$EF_{BM,y}$	$EF_{CM,y}$	Source
Value/Unit	1.1293tCO_2e/MWh	0.7242tCO_2e/MWh	1.0280tCO_2e/MWh	Notification on

Electricity grid included in NECPG	Liaoning Power Grid, Jilin Power Grid, Heilongjiang Power Grid	Determining Baseline Emission Factor of China's Grid[2]

(b) Technology to be employed by the Project:

The project scenario is the implementation of the proposed project, the installation and operation of 58 sets of wind turbines with a total capacity of 49.3MW which will supply an average annual generation of 120,069MWh to NECPG and replace the same amount of electricity generated by fossil fuel fired power plants connected to NECPG. Each wind turbine will have a box transformer which transforms the voltage to 35kV from which the 35kV line will link into the on-site 110kV switchgear at the project site. By the 110kV line, the electricity generated by the proposed project is delivered to the power grid. The wind turbines and transmission facility could be monitored and controlled by onsite central control room.

The technologies employed in the proposed project activity are advanced domestic technologies, which is no technology transfer activity involved. The main advantage of this domestic manufactured turbine is its' adaptability to different types of wind resources and its improved generation efficiency, so it has been widely installed in China.

[2] Notification on Determining Baseline Emission Factor of China's Grid issued by China's DNA on July 2nd,2009 http://cdm.ccchina.gov.cn

Table 1 Main technical information of wind turbines used in the proposed project.

Parameter	Value	Source
Installed Capacity（kW）	850	Feasibility Study Report and wind turbine purchasing contract
Diameter of rotor（m）	G58-850/G52-850	
Rated wind speed（m/s）	13	
Cut-in wind speed（m/s）	3	
Cut-out wind speed（m/s）	21	
Survival wind speed（m/s）	37.5	
Swept area (m^2)	2642/2214	
Lifetime	20 year	
Generator	Doubly fed asynchronous generator	

The plant load factor of the proposed project is 27.80% determined by a third qualified party contracted by the project participant (e.g. China Fulin Windpower Development Corporation) and the plant load factor was provided to the government while applying the project activity for implementation approval.

The technologies employed in the proposed project activity are advanced domestic technologies, which is no technology transfer activity involved.

(c) The baseline scenario is the same as the scenario existing prior to the start of implementation of the project activity, which will supply an average annual generation of 120,069MWh to NECPG and replace the same amount of electricity generated by fossil fuel fired power plants connected to NECPG to reduce greenhouse gas emissions of CO_2.

A.4.4. Estimated amount of emission reductions over the chosen crediting period:

The renewable crediting period is adopted by the Project. It is expected that the Project will generate emission reductions for about 123,430tCO_2e per year over the first 7-year crediting period from Mar. 1st, 2012 to Feb. 28th, 2019.

Years	Annual estimation of emission reductions in tonnes of CO_2e
01/03/2012-31/12/2012	102,859
2013	123,430
2014	123,430
2015	123,430
2016	123,430
2017	123,430
2018	123,430
01/01/2019-28/02/2019	20,571
Total estimated reductions (tonnes of CO_2e)	864,010
Total number of crediting years	7
Annual average over the crediting period of estimated reductions (tonnes of CO_2e)	123,430

A.4.5. Public funding of the project activity:

No public funds from countries in Annex I are involved in the proposed project.

SECTION B. Application of a baseline and monitoring methodology:

B.1. Title and reference of the approved baseline and monitoring methodology applied to the project activity:

The Project applies the approved baseline and monitoring methodology ACM0002-Consolidated methodology for grid-connected electricity generation from renewable sources (Version 12.1.0) and refers to *Tool for the demonstration and assessment of additionality* (Version 05.2.1) and *Tool to calculate the emission factor for an electricity system* (Version 02.2.1).

For more information please refer to following link:

http://cdm.unfccc.int/methodologies/PAmethodologies/approved.html

B.2. Justification of the choice of the methodology and why it is applicable to the project activity:

The Project meets the applicable criteria of the methodology ACM0002 (Version 12.1.0) due to following reason:

The Project is a grid-connected renewable power generation project activities that installs a new wind power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant).

Therefore the methodology ACM0002 (Version 12.1.0) is applicable to the Project.

B.3. Description of the sources and gases included in the project boundary:

The proposed project is the installation of a new grid-connected renewable power plant, and the baseline scenario is the following:

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".

The spatial extent of the project boundary includes the proposed project and all power plants connected physically to the **Northeast China Power Grid** that the CDM project power plant is connected to. **Northeast China Power Grid** is defined as the project electricity system. In accordance with the boundary definitions of the Chinese DNA[3], Northeast China Power Grid consists of independent province-level electricity systems including the Liaoning, Jilin and Heilongjiang province that can be dispatched without significant transmission constraints. The proposed project located in the central eastern part of Heilongjiang province is connected into

[3] http://qhs.ndrc.gov.cn/qjfzjz/t20090703_289357.htm

Northeast China Power Grid by Heilongjiang province Grid. The Northeast China Power Grid is net electricity export grid.

The spatial extent of the project boundary includes the project site and all power plants connected physically to NECPG.

The sources and gases included in the project boundary are listed as below:

	Source	**Gas**	**Included?**	**Justification / Explanation**
Baseline	CO2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity.	CO_2	Yes	Main emission sources
		CH_4	No	Minor emission source
		N_2O	No	Minor emission source
Project Activity	The proposed project is a renewable power source	CO_2	No	According to ACM0002, the project emission of wind power project activity is not considered.
		CH_4	No	According to ACM0002, the project emission of wind power project activity is not considered.
		N_2O	No	According to ACM0002, the project emission of wind power project activity is not considered.

The Project boundary can be explained by the following flow diagram, figure 3:

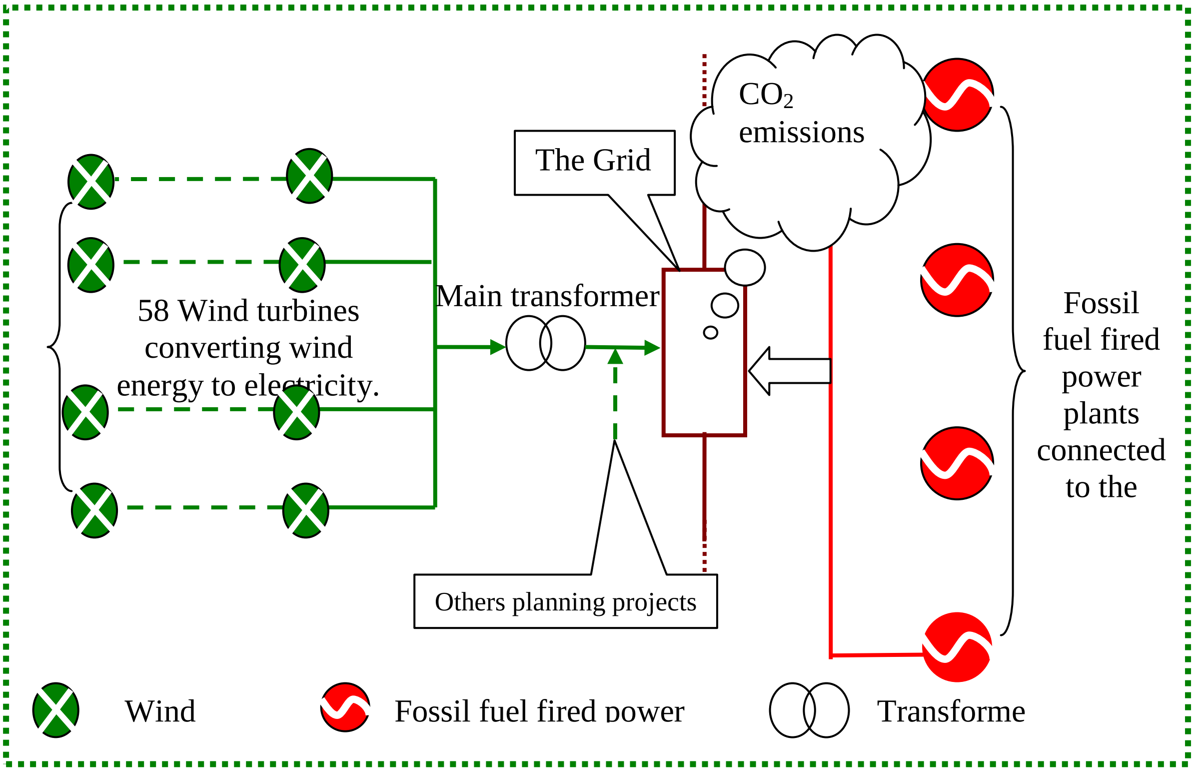

Figure 3 the flow diagram of the project

B.4. Description of how the baseline scenario is identified and description of the identified baseline scenario:

The proposed project is the installation of a new grid-connected renewable power plant and the baseline scenario is the following as per ACM0002 (Version 12.1.0):

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".

The proposed project is connected to Heilongjiang Province Grid, which is an integrated part of Northeast China Power Grid (NECPG). Therefore, the baseline scenario of the proposed project is provision of equivalent amount of annual power output by the NECPG where the proposed project is connected, which is the continued operation of the existing power plants and the addition of new generation sources on the NECPG to meet the electricity demand.

According to ACM0002, baseline emissions are equal to net electricity generation that is produced and fed into the NECPG, multiplied by the baseline emission factor. The baseline emission factor (EF_y) is calculated as a Combined Margin (CM). The analysis and description in B.5 and B.6 will support the baseline scenario selected above.

B.5. Description of how the anthropogenic emissions of GHG by sources are reduced below those that would have occurred in the absence of the registered CDM project activity (assessment and demonstration of additionality):

The CDM incentive was seriously considered in the Feasibility Study Report of the proposed project, which clearly states that "The IRR of the proposed project is lower than the benchmark, to ensure a successful implementation of the proposed project, the project developer should apply the CDM instrument according to the related regulation of the State, and improve the project return through GHG emission reduction trading". Besides, in order to secure the successful implementation of the proposed project, the project owner signed a consultant contract to start CDM development and conducted negotiation with the potential buyers in parallel with its implementation.

The project activity is not the baseline scenario and possesses additionality, which is demonstrated below in a step-wise manner using the latest version (05.2.1) of "Tool for demonstration and assessment of additionality".

The CDM had been taken into serious consideration by the project owner during the investment decision of the proposed project, as can be clearly shown in the following Timeline of the Proposed Project.

Table 2 the timeline of the project

No.	Timeline	Milestone
1	19/05/2008	EIA finished
2	25/07/2008	EIA approval
3	09/2008	Feasibility study finished
4	24/03/2009	Approval of the FSR issued by DRC of Heilongjiang Province
5	02/04/2009	The directorate of project owner thought that the potential CERs revenue can help the project to be financial attractive and decided to invest the construction of the project.
6	04/2009	The CDM Consultation Contact signed.
7	05/2009	The questionnaires made.
8	06/05/2009	The stakeholder symposiums hold.
9	12/05/2009	When the wind turbine purchasing contract signed
10	14/05/2009	When the transformer purchasing contact signed
11	04/06/2009	The construction period started.
12	05/06/2009	When the construction contract signed
13	02/07/2009	The prior Consideration of the CDM Form of the project has sent to China DNA
14	10/07/2009	The Validation Contact signed with BV.
15	22/07/2009	The confirm letter of EB has received and the information on the project has been uploaded in EB website.
16	19/08/2009	The confirm letter from China DNA has received.
17	12/09/2009	The Project is public available on the UNFCCC CDM website

The additionality of the Project is demonstrated by using the *Tool for the Demonstration and Assessment of Additionality* (Version 05.2.1) approved by the CDM EB and requested by the methodology ACM0002 (Version12.1.0). The *Tool for the Demonstration and Assessment of Additionality* (Version05.2.1) provides for a step-wise approach to demonstrate and assess the additionality. These steps include:

Step 1. Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a. Define alternatives to the project activity.

For the Project, the possible alternative scenarios in absence of the CDM project activity should be as follows:

Alternative I: To implement the proposed project activity, but not as a CDM project activity;

Alternative II: To construct a thermal power plant with the same electricity output as the Project;

Alternative III: To construct a power plant using other renewable resources with the same electricity output as the Project;

Alternative IV: To provide for the same annual electricity output as the Project by NECPG.

For Alternative III, besides wind energy, other kinds of energy like solar PV, geothermal, biomass and hydro are the possible grid-connected renewable energy technologies that could be applied in China.

For the alternative c), besides wind energy, other kinds of energy like solar PV, geothermal, biomass and hydro are the possible grid-connected renewable energy technologies that could be applied in China. Due to different grid connected situation, high operating cost and less of national policies[4], solar PV power project is less attractive for investor in China. For geothermal power project, due to limited thermal resources in China which has great thermal resources sited in Tibet and Taiwan[5] and geothermal power project is less attractive for investor without support from the national policies[6]. According to *the Distributing and utilization of geothermal generation resource in China*, geothermal generation resource is not suitable to build power plant in Heilongjiang[7]. Biomass power project is also not suitable choice for development because of high cost, low payoff and less support from the national policies[8]. Therefore, the possible grid-connected renewable energy technologies such as solar PV, geothermal, biomass with the same annual electricity output as the proposed project are alternatives far from being attractive investment in the grid in China. This region is lack of water resource and water loss and soil erosion is serious[9], which is not suitable to build hydropower station. Therefore, though the alternative c) is in compliance with all mandatory laws and regulations, is not a realistic alternative.

Therefore, though Alternative III is in compliance with all mandatory laws and regulations, is not a realistic alternative.

To sum up, Alternative I, Alternative II and Alternative IV should be considered in the following analysis.

Sub-step 1b. Enforcement of applicable laws and regulations:

Alternative I: To implement the proposed project activity, but not as a CDM project activity. The alternative is in compliance with current laws and regulations of China.

Alternative II: According to the applicable laws and regulations, the alternative b) should be eliminated from the following consideration because it does not comply with the national regulation for controlling small scale thermal power plant. In 2009, the average operation time of thermal power plant in Heilongjiang Province is 4257hr[10], and the effective operation time of

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4 http://www.wefweb.com/news/200916/1413073809_0.shtml

http://www.newenergy.org.cn/html/0105/5311032740_1.html

5 http://www.newenergycn.com/drn_news.asp?id=125

6 http://app.chinamining.com.cn/Newspaper/E_Mining_News_2010/2010-08-05/1280970747d43096.html

7 http://www.newenergycn.com/drn_news.asp?id=125

http://www.newenergy.org.cn/Html/0097/780928486.html

8 http://www.ce.cn/cysc/ny/xny/200901/12/t20090112_17928711.shtml

9 http://finance.eastmoney.com/news/1350,20110528138824874.html

10 China Electric Power Yearbook 2010

proposed project is 2364.6hr[11]. To provide the same output as the proposed project, the alternative thermal power plant will has the capacity of 27.38MW. Then it will be categorized as the small scale thermal power plant and should be shut down according to the regulations from NDRC[12]. According to this regulation, the thermal power plant that has the capacity of less than 50MW should be shut down and the construction of thermal power plant that has the capacity of less than 135MW will be forbidden within the grid connected area. Therefore, Alternative II conflicts with Chinese regulations.

Alternative IV: To provide for the same annual electricity output as the Project by NECPG. The alternative is in compliance with current laws and regulations of China.

To sum up, Alterative I and Alterative IV should be considered in the following analysis.

Step 2. Investment analysis

Sub-step 2a. Determine appropriate analysis method

Tool for the Demonstration and Assessment of Additionality (Version05.2.1) provides three analysis methods to apply for the investment analysis: the simple cost analysis (option I), the investment comparison analysis (option II) and the benchmark analysis (option III).

For the proposed project, the simple cost analysis method is not applicable because the project activity will generate economic benefit (from electricity sale) other than CDM related income. The investment comparison analysis method is also not applicable because the Alterative IV is the Northeast China Power Grid rather than a new investment project.

To conclude, the proposed project will use the benchmark analysis method based on total investment IRR to identify whether or not that the financial indicators of the proposed project is lower than relevant benchmark value.

Sub-step 2b – Option III. Apply benchmark analysis

In according with *Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects* issued by former State Power Corporation of China[13], the financial internal rate of return (Project IRR) for total investment as benchmark in China's power generation industry is 8%(after tax), considering economic assessments of hydropower projects, fossil fuel fired projects, transmission and substation projects. Nowadays China's existing wind power projects have also applied it as the benchmark IRR. Therefore, 8% is adopted as the financial benchmark IRR for the Project. If the total investment's IRR of the Project is less than 8%, the Project will be financially unfeasible and then be additional.

Sub-step 2c. Calculation and comparison of financial indicators:

Basic parameters of the Project

The basic parameters to calculate the financial indicators of the Project are listed in Table 3.

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[11] The Feasible Study Report of proposed project

[12] On Prohibition of 135MW and Smaller-scale Coal-fired Power *Plants, General Office of State Council(*http://www.gov.cn/gongbao/content/2002/content_61480.htm*)*

[13] http://www.law-lib.com/law/law_view1.asp?id=8867

Table3. Basic parameters for IRR calculation[14]

Items	Unit	Data	Source
Installed Capacity	MW	49.3	Feasibility Study Report
Estimated annual electricity generated	MWh	120,069	Feasibility Study Report
Project lifetime	year	20	Feasibility Study Report
Total static investment	Million Yuan	482.69	Feasibility Study Report
Electricity tariff(incl. VAT) (<30000hours)	Yuan/kWh	0.6100	Feasibility Study Report
Electricity tariff(incl. VAT) (>30000hours)	Yuan/kWh	0.4100	Feasibility Study Report
VAT	%	8.5[15]	Feasibility Study Report
Income tax	%	25	Feasibility Study Report
Tax of expense for city maintenance and construction	%	0	Feasibility Study Report
Tax of education fee addition	%	0	Feasibility Study Report
Period of depreciation	years	15	Feasibility Study Report
Rate of depreciation	%	6.33	Feasibility Study Report
Interest rate	%	5.346	Loan letter
Equity-dept rate	%	75:25	Loan letter
Annual O&M cost	Million Yuan	14.0562	Feasibility Study Report
Rate of fixed assets maintenance	**%**	1.5	Feasibility Study Report
Material cost	Yuan/kW	20	Feasibility Study Report

14 According to Interim Rules on Economic Assessment of Electrical Engineering Retrofit Project, the fixed value should be used in the investment analysis.

15 According to the current law in China, the Statute of People's Republic of China, the value added tax is defined as 17% for wind power industry. In order to encourage the development of wind powers in China, the Ministry of Finance and State Administration of Taxation provides a tax reduction for wind power projects which is 8.5%. It complies with the Notice of Value added Tax Policy Regarding Products Using Certain Synthesized Resources and Other Products [2001] No. 198 issued by the Ministry of Finance and the State Administration of Taxation in Dec. 2001 at the time of FSR was finished.

Miscellaneous cost	Yuan/kW	50	Feasibility Study Report
No. of employee		20	Feasibility Study Report
Annual salary per capita	Million Yuan	0.05	Feasibility Study Report
Rate of material benefits	%	14	Feasibility Study Report
Rate of labor insurance	%	17	Feasibility Study Report
Rate of house fund	%	10	Feasibility Study Report
Staff and workers' bonus and welfare fund	%	0	Feasibility Study Report

Comparison of financial indicator

Based on these data listed on Table 3 above, the IRR of the total investment of the Project is only 6.81% that is lower than the benchmark IRR of 8%.

Taking into account the income from CERs (10.5 €/tCO_2e), the total investment's IRR of the Project will be 10.39%, which will higher than the benchmark IRR of 8%. Therefore, the Project is economically attractive, which means that the CDM revenues could help the Project overcome the investment barrier.

Table 4 Comparison of financial indicators with and without income from CERs

Items	Unit	Without income from CERs	Benchmark	With income from CERs
IRR	%	6.81	8	10.39

Sub-step 2d. Sensitivity analysis

The purpose of the sensitivity analysis is to examine whether the conclusion regarding the financial viability of the proposed project is sound and tenable with those reasonable variations in the assumptions.

Four factors are considered in following sensitivity analysis:

1) Total static investment
2) Tariff
3) Estimated annual output
4) Annual operation and maintenance cost(**O&M cost**)

The four financial parameters were identified as the main variable factors for sensitive analysis of financial attractiveness. Assuming that the above four factors fluctuate within the range of -10%~+10%[16], the corresponding impacts on the IRR of total investment of the Project are shown in Table5-8 and Figure 4.

[16] Based on the information from the Project owner, the actual total investment exceeds the estimated investment in *FSR* due to the increasing price of raw material and labour force. Meanwhile, the operation cost of the Project will also exceed the estimation in *FSR*. Consequently assuming the variation range for total investment and annual O&M

Table5 Sensitivity of total investment IRR to Total static investment

Variation (%)	-10.0	-7.8	-5.0	-2.5	0	2.5	5.0	7.5	10.0
IRR (%)	8.37	8.00	7.55	7.17	6.81	6.45	6.12	5.79	5.48

Table6 Sensitivity of total investment IRR to O&M cost

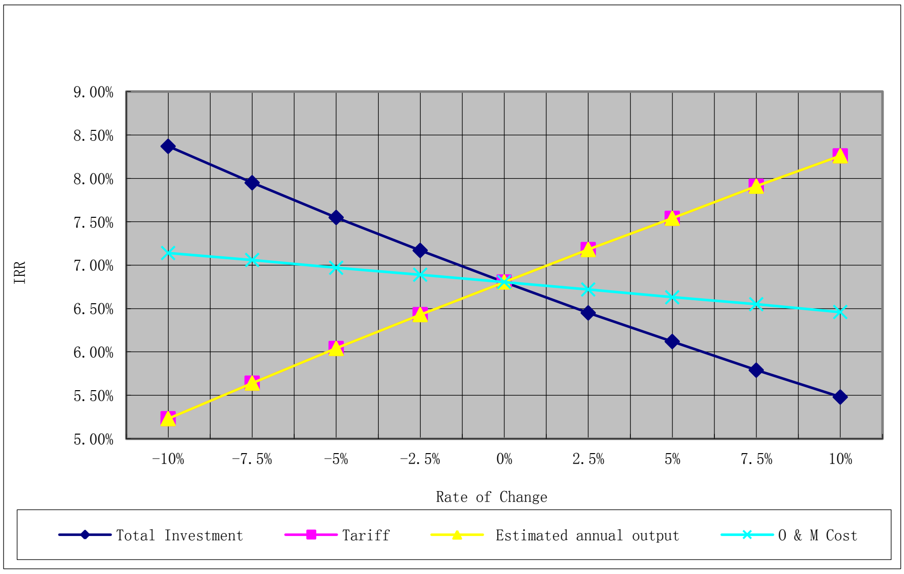
Variation (%)	-36.3	-10	-7.5	-5.0	-2.5	0	2.5	5.0	7.5	10.0
IRR(%)	8.00	7.14	7.06	6.97	6.89	6.81	6.72	6.63	6.55	6.46

Table7 Sensitivity of total investment IRR to Estimated annual output

Variation (%)	-10.0	-7.5	-5.0	-2.5	0	2.5	5.0	8.18	10.0
IRR (%)	5.23	5.64	6.04	6.43	6.81	7.18	7.54	8.00	8.26

Table8 Sensitivity of total investment IRR to Tariff

Variation (%)	-10.0	-7.5	-5.0	-2.5	0	2.5	5.0	8.18	10.0
IRR (%)	5.23	5.64	6.04	6.43	6.81	7.18	7.54	8.00	8.26



cost as ±10% is conservative. The electricity generation is calculated by China Fulin Windpower Development Corporation based on wind sources data of 37 years from 1971 to 2007 and in general will be very close to real situation. Therefore, ±10% can be considered logical and reasonable for electricity generation. The tariff is primarily ratified in the general price of location. Therefore, it's unlikely to vary much and variation scope within the range of 10% is logical and reasonable.

Figure 4 IRR of total investment sensitivity analysis

In the case that total investment decreases by about 7.80%, the IRR of the proposed project begins to exceed the benchmark. Considering the majority of total investment is used to buy wind turbines, whose price has been determined by signing the equipment purchase agreement (EPA). For the proposed project, the static total investment was presumed to be 482.69 million RMB in the FSR, in which the cost for the wind turbines was of 281.01 Million Yuan about 58.22% of the total static investment in the FSR. However, the actual price is 306.30 million Yuan in the contract, about 9.00% higher than the value in the FSR. The rest of the static total investment is for installation of the wind turbines and the construction of the wind turbine foundation and some other buildings, due to the increasing trend of construction material costs[17], is also unrealistic to be lowered down[18] [19].Hence within the reasonable range of total investment, the proposed project is always lack of financial attractiveness.

In the case that the expected power tariff increases by about 8.18%, the IRR of the proposed project begins to exceed the benchmark. However, the tariff of wind power plants in Heilongjiang Province in 2007 and 2008 has been approved by NDRC[20], which is 0.5622 Yuan/kWh (Excluding VAT)[21] [22]in the first 30000 hours. Considering the fact that the guiding tariff has been determined, it is almost unlikely for this tariff of 0.6100 Yuan/kWh (including VAT)[23] [24]in the first 30000 full load hours has an increase. However, the tariffs after 30000 full load hours implemented the average tariff of the local grid. Since Heilongjiang Power Grid is dominated by the thermal power plants which supplies more than 95% of the total power generation of that province in 2008[25], the average tariff in the grid is mainly determined by thermal power tariff. Therefore, the estimated tariff of 0.4100 Yuan/kWh (including VAT) applied after 30000 full load hours was more conservative compared with the average tariff of 0.3655 Yuan/kWh (including VAT) [26]in Year 2008. Furthermore, as per "*Information note on the highest tariffs applied by the EB in its decisions on registration of projects in the People's Republic of China (version 2.0)*"[27], the proposed project is still additional with the highest tariff of Heilongjiang. As a result, the proposed project is always lack of financial attractiveness because the tariff would not increase by 8.18%.

In the case that the estimated annual output increases by about 8.18%, the IRR of the proposed project begins to exceed the benchmark. According to the FSR, the Average annual output is

17 http://finance1.jrj.com.cn/news/2008-03-28/000003465715.html
18 http://news.hexun.com/2008-07-01/107093401.html
19 http://news.chinanewenergy.com/html/20081007/20550.html
20 The National Development & Reform Commission
21 http://jgs.ndrc.gov.cn/zcfg/t20080218_193011.htm
22 http://www.ndrc.gov.cn/zcfb/zcfbtz/2008tongzhi/t20080813_230718.htm
23 http://jgs.ndrc.gov.cn/zcfg/t20080218_193011.htm
24 http://www.ndrc.gov.cn/zcfb/zcfbtz/2008tongzhi/t20080813_230718.htm

25 China Electric Power Yearbook 2009

26 Notice on adjustment of the tariff in NECPG issued by NDRC in 2008

27 Para 53, EB 53, http://cdm.unfccc.int/Reference/Notes/reg_note07.pdf

estimated basing on the 30-year wind statistic data provided by local meteorological station. As shown in the 30-year wind statistic data in FSR, the average wind speed of 30-year is 3.01m/s, the average wind speed of 20-year is 2.93m/s, and the average wind speed of 10-year is 2.86 m/s. Moreover, the wind speed is decreasing and trending to be stable during recent 10 years[28].Moreover, the annual power generation was estimated based on the data of the average wind speed of 30-year of 3.01m/s that was 5.24% higher than the averaged wind speed of 10-year. The calculation of local wind resources is based on the more than 30 years monitoring data by local weather department. Therefore, the probability that electricity outputs is 8.18% higher than the estimated value is unreasonable.

The impact of the annual O&M cost is the slightest, as the IRR of the proposed project begins to exceed the benchmark when the annual O&M cost decreases by 36.30%. However, according to the Economical Assessment Method and Parameters for Construction Project (3rd edition) issued by NDRC and Ministry of Construction of China, the detailed operation costs is composed of Maintenance Fee, Salary and Welfare Fee, Insurance Fee, etc. In the recent years, the price of material and salaries of the employees are gradually increasing in China[29]. At the same time, the wind turbines demand exceeds supply in the whole world[30] at present, which leads to the price increasing and the shortage of accessorial equipments of wind turbines. Under this condition, the maintenance costs for accessorial equipments of wind turbines are also increasing during the operation period. As above description, the annual O&M cost is gradually increasing during the operation period for the proposed project. Moreover, according to the document "Wind Energy Operations & Maintenance Report[31]", $0.027/kWh is the average value of O&M costs obtained from report surveys. Based on the figure mentioned, the annual O&M costs of the proposed project is calculated as 2269.3(10000Yuan)[32] that is higher than the average annual O&M costs of 1406 (10000Yuan) in the IRR calculation worksheet. Therefore, the proposed project is always lack of financial attractiveness within the reasonable range of annual O&M cost.

Step3. Barrier analysis

No barrier analysis has been applied.

Step 4 Common practice analysis

Sub-step 4a. Analyze other activities similar to the proposed project activity

In 2009[33], wind power accounted for only 7.30% of the total installed capacity of the Northeast China Power Grid and 2.77% of actual supply on the Northeast China Power Grid[34], it is clear

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28 The Feasibility Study Report of the proposed project.

29 http://news.iyaxin.com/content/2010-11/16/content_2303246.htm

30 http://energy.people.com.cn/GB/5720709.html

31 http://www.wind-watch.org/documents/wind-energy-operations-maintenance-report/

32 The calculated annual O&M costs of the proposed project = $0.027/kWh*120069MWh ($EG_y$)*7(USD:RMB)

33 China Electric Power Yearbook 2010

34 China Electric Power Yearbook 2010

that wind power is not a common practice generally.
According to the definitions of other activities similar to the proposed project activity in "Tool for the Demonstration and Assessment of Additionality", the similar projects are considered similar if they are in the same country/region and/or rely on a broadly similar technology, are of a similar scale, and take place in a comparable environment with respect to regulatory framework, investment climate, access to technology, access to financing, etc. Therefore, the similar activities are defined as the projects operated after 2002 in Heilongjiang Province.

The activities similar to the proposed project activity were selected according to the following three rules:

1. The activities should locate in the same province. Firstly, the wind power project was implemented under the administration of provincial level government. Secondly, the activities in the same province have the similar wind resource, grid structure, geological and economic developing conditions. The former two factors affected the estimated average annual output and the later affected the total investment respectively. Consequently, the similar activities should be limited within Heilongjiang Province which the proposed projects located.
2 . The similar activities should be implemented after 2002. Before 2002, Chinese Power System had not been reformed and power grid and power generation had not been separated. There was no bidding process was required and power companies and grid companies share the same interests. The Power System reform happened in 2002 in China. As a result, new power project, including wind power, was expected to compete under more commercial conditions. Moreover, the Chinese Government launched the Wind Bidding Mechanics. Accordingly, the wind power projects were developed in the different regulatory condition after 2002.
3. Following the clarification made in the EB38 meeting report (paragraph 60), project activities which have been registered or validated as CDM projects on the UNFCCC CDM website are not identified as similar activities to the proposed project.

According to the terms above mentioned and the statics of wind power project in China by Shi pengfei in 2007, 2008, 2009 and 2010, the similar projects with the proposed project in Heilongjiang province since 2002 are listed as follows:

Table 9 Wind Farms in Heilongjiang Province[35] [36] [37]

Project Title	Commissioning Date	Capacity (MW)	Note
Huafu Mulan Wind Farm	2003.12	12 MW	Demonstration Project

[35] Shi Pengfei (Deputy Director, Chinese Wind Energy Association), Statistics on China Wind Farm Installed Capacity in 2007.
Shi Pengfei (Deputy Director, Chinese Wind Energy Association), Statistics on China Wind Farm Installed Capacity in 2008.
Installed capacity statistics of China wind power by Chinese Wind Energy Association (CWEA),2009
[36] http://cdm.unfccc.int/index.html
[37] http://cdm.ccchina.gov.cn/web/index.asp

Haufu Fujin Wind Farm	2004.09	24.3 MW	Demonstration Project

Sub-step 4b. Discuss any similar options that are occurring:

The existing wind farm projects do not call into question the claim that the proposed project is financially unattractive as discussed in Step 2. The two wind farms in Table 3 benefited from more favorable tariff and were funded by international low interest loan[38] or national soft loan[39], while the proposed project does not enjoy these favourable policies. The other projects similar to the proposed project activity in Heilongjiang province have been applied for being as CDM projects; some of them have been registered, the rest have been approved by Chinese government and are under validation[40]. Since the proposed project has no comparable financial or the tariff advantage as these projects, the proposed project fulfils the requirement of additionality.

In conclusion, the proposed project activity passed all criteria of "Tool for the demonstration and assessment of additionality (Version 5.2.1)". The proposed project is additional.

B.6. Emission reductions:

B.6.1. Explanation of methodological choices:

The proposed project is the installation of a new grid-connected renewable power plant, and the baseline scenario is the following:

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".

Baseline emissions

Baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$
(1)

Where:

[38] http://www.newenergy.org.cn/Html/9991/19991799.html
[39] http://www.chinapower.com.cn/newsarticle/1005/new1005504.asp
[40] http://cdm.ccchina.gov.cn/web/index.asp

BE_y = Baseline emissions in year *y* (tCO_2/yr).
$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr).
$EF_{grid,CM,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year *y* calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO_2/ MWh).

Since the proposed project is the installation of a new grid-connected renewable power plant, then $EG_{PJ,y}$ is calculated using the following formula:

$$EG_{PJ,y} = EG_{facility,y}$$

(2)

Where:
$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr).
$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr).

$$EG_{facility,y} = EG_{\text{export},y} - EG_{import,y}$$

(3)

Where:
$EG_{\text{export},y}$ = Quantity of electricity generation exported by the proposed project to the grid in year y (MWh/yr).
$EG_{import,y}$ = Quantity of electricity generation imported by the proposed project from the grid in year y (MWh/yr).

The methodological tool "Tool to calculate the emission factor for an electricity system" (version 02.2.1) determines the CO_2 emission factor for the displacement of electricity generated by power plants in an electricity system, by calculating the "combined margin" emission factor (CM) of the electricity system. The CM is the result of a weighted average of two emission factors pertaining to the electricity system: the "operating margin" (OM) and the "building margin" (BM). The operating margin is the emission factor that refers to the group of existing power plants whose current electricity generation would be affected by the proposed CDM project activity. The build margin is the emission factor that refers to the group of power plants whose construction and future operation would be affected by the proposed CDM project activity.

The methodological tool "Tool to calculate the emission factor for an electricity system" (version 02.2.1) provides procedures to determine the following parameters:

Parameter	**SI Unit**	**Description**

$EF_{grid,CM,y}$	tCO_2/MWh	Combined margin CO_2 emission factor for the project electricity system in year y
$EF_{grid,BM,y}$	tCO_2/MWh	Build margin CO_2 emission factor for the project electricity system in year y
$EF_{grid,OM,y}$	tCO_2/MWh	Operating margin CO_2 emission factor for the project electricity system in year y

The following seven steps are applied to calculate the emission factor for an electricity system:

Step1. Identify the relevant electricity systems;
Step2 Choose whether to include off-grid power plants in the project electricity system (optional);
Step3.Select a method to determine the operating margin (OM);
Step4. Calculate the operating margin emission factor according to the selected method;
Step5. Calculate the build margin (BM) emission factor;
Step6. Calculate the combined margin (CM) emissions factor.

Step1: Identify the relevant electricity systems

Using the boundary definitions of the Chinese DNA[41], the spatial extent of the project boundary includes Proposed Project and all power plants connected physically to the Northeast China Power Grid that the CDM project power plant is connected to Northeast China Power Grid is defined as the project electricity system, which consists of independent province-level electricity systems including Liaoning, Jilin and Heilongjiang province that can be dispatched without significant transmission constraints.

Electricity transfers from connected electricity systems to the project electricity system are defined as **electricity imports** and electricity transfers to connected electricity systems are defined as **electricity exports**.Since the Northeast China Power Grid has no electricity import, the spatial extent is limited to the project electricity system (Northeast China Power Grid).

Step2: Choose whether to include off-grid power plants in the project electricity system (optional)

Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.
Option II: Both grid pants and off-grid power plants are included in the calculation.

The proposed project calculation only included grid power plants. Therefore, Option I is selected.

[41] http://cdm.ccchina.gov.cn/WebSite/CDM/UpFile/File2333.pdf

Step3: Select a method to determine an operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods:

(a) Simple OM, or
(b) Simple adjusted OM, or
(c) Dispatch data analysis OM, or
(d) Average OM

The simple OM method (option a) can only be used if low-cost/must-run resources [42] constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production.

The dispatch data analysis (option d) cannot be used if off-grid power plants are included in the project electricity system as per Step 2 above.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

- *Ex ante* option: If the *ex ante* option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the CDM-PDD to the DOE for validation. For off-grid power plants, use a single calendar year within the 5 most recent calendar years prior to the time of submission of the CDM-PDD for validation.

- Ex post option: If the *ex post* option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring. If the data required to calculate the emission factor for year y is usually only available later than six months after the end of year y, alternatively the emission factor of the previous year $y-1$ may be used. If the data is usually only available 18 months after the end of year y, the emission factor of the year proceeding the previous year $y-2$ may be used. The same data vintage (y , $y-1$ or $y-2$) should be used throughout all crediting periods.

For the dispatch data analysis OM, use the year in which the project activity displaces grid electricity and update the emission factor annually during monitoring.

[42] Low-cost/must-run resources are defined as power plants with low marginal generation costs or power plants that are dispatched independently of the daily or seasonal load of the grid. They typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation. If coal is obviously used as must-run, it should also be included in this list, i.e. excluded from the set of plants.

The data vintage chosen should be documented in the CDM-PDD and not be changed during the crediting periods.

Power plants registered as CDM project activities should be included in the sample group that is used to calculate the operating margin if the criteria for including the power source in the sample group apply.

The justifications of the choice of method to calculate OM emission factor are as follows.

Method (c): The dispatch data analysis OM emission factor is determined based on the power units that are actually dispatched at the margin during each hour *h* where the project is displacing electricity. This method requires the dispatch order of each power plant and the dispatched electricity generation of all the power plants in the power grid during each hour. Since the dispatch data, power plants operation data are considered as confidential materials and only for internal usage and are not available publicly. Thus, method (c) is not applicable for the proposed project.

Method (b): Method (b) requires the annual load duration curve of the power grid and the load data of every hour data during the whole year on the basis of the time order. As mentioned above, the dispatch data and detailed load curve data are not available publicly. Therefore, method (b) is not applicable for the proposed project as well.

In terms of Method (d) and Method (a): The average OM emission factor (option d) is calculated as the average emission rate of all power plants serving the grid, using the methodological guidance as described under (a) above for the simple OM, but including in all equations also low-cost/must-run power plants. The simple OM method (option a) can only be used if low-cost/must-run resources constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production. Low operating cost and must run resources typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation. If coal is obviously used as must-run, it should also be included in this list, i.e. excluded from the set of plants. Considering the low-cost/ must run resources only constitute 4.72%, 6.45% ,7.98%,5.25% and 5.24%of total generation of Northeast China Power Grid from the year 2003 to 2007, respectively (China Electric Power Yearbooks 2004-2008).. Therefore, method (a) is chosen to calculate OM emission factor for the proposed project.

In conclusion, the Ex ante option of the data vintages is chosen to calculate the emission factor of the Northeast China Power Grid by using the simple OM method (option a) for the proposed project.

Step 4: Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated:

Option A: Based on the net electricity generation and a CO_2 emission factor of each power unit[43]; or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Option B can only be used if:

(a) The necessary data for Option A is not available; and

(b) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(c) Off-grid power plants are not included in the calculation (i.e., if Option I has been chosen in Step 2).

For the proposed project, the data on fuel consumption, net electricity generation and the average efficiency of each power unit are unavailable, thus option A cannot be used. Nevertheless, the data on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system are available, nuclear and renewable power generation are considered as low-cost / must-run power sources and the quantity of electricity supplied to the grid by these sources is known, and, off-grid power plants are not included in the calculation which means Option I has been chosen in Step 2, therefore, Option B can be used.

On Option B, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_i \left(FC_{i,y} \times NCV_{i,y} \times EF_{CO_2,i,y}\right)}{EG_y}$$

(4)

Where:

$EF_{grid,OMsimple,y}$ = Simple operating margin CO_2 emission factor in year y (tCO_2/MWh)

$FC_{i,y}$ = Amount of fossil fuel type i consumed in the project electricity system in year y (mass or volume unit)

$NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i in year y (GJ/mass or volume unit)

$EF_{CO_2,i,y}$ = CO_2 emission factor of fossil fuel type i in year y (tCO_2/GJ)

EG_y = Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants / units, in year y (MWh)

[43] Power units should be considered if some of the power units at the site of the power plant are low-cost/must-run units and some are not. Power plants can be considered if *all* power units at the site of the power plant belong to the group of low-cost/must-run units or if *all* power units at the site of the power plant do *not* belong to the group of low-cost/must-run units.

i = All fossil fuel types combusted in power sources in the project electricity system in year y

y = The relevant year as per the data vintage chosen in Step 3

For this approach (simple OM) to calculate the operating margin, the subscript m refers to the power plants/units delivering electricity to the grid, not including low-cost/must-run power plants/units, and including electricity imports[44] to the grid. Electricity imports should be treated as one power plant m.

Regarding parameter selection, local values of $NCV_{i,y}$ and $EF_{CO_2,i,y}$ should be used where available. If no such values are available, IPCC world-wide default values are preferable. The Net Calorific Value ($NCV_{i,y}$) of each type of fossil fuel used in the calculation comes from China Energy Statistic Yearbook 2008. Emission factors ($EF_{CO_2,i,y}$) of each type of fossil fuel come from IPCC 2006 default values.

As chosen in step 3, the simple OM emission factor is calculated by using *ex ante* option of data vintages, i.e. a 3-year generation-weighted average, based on the most recent data available at the time of submission of the CDM-PDD to the DOE for validation. And the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

The data of installed capacity, electricity generation and fuel consumptions are all from China Energy Statistical Yearbooks 2006-2008 and China Electric Power Yearbooks 2006-2008.

Given the above, the simple operating margin CO_2 emission factor ($EF_{grid,OMsimple,y}$) of Northeast China Power Grid is **1.1293 tCO_2/MWh.** The detailed calculations and data are listed in the annex 3 (The baseline emission factor OM is same as that provided by Chinese DNA, the website is
(http://cdm.ccchina.gov.cn/WebSite/CDM/UpFile/File2333.pdf).

Step 5: Calculate the build margin (BM) emission factor

The sample group of power units m used to calculate the build margin consists of either[45]:

(a) The set of five power units that have been built most recently, or
(b) The set of power capacity additions in the electricity system that comprise 20% of the system generation (in MWh) and that have been built most recently[46].

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[44] As described above, an import from a connected electricity system should be considered as one power source.

[45] If this approach does not reasonably reflect the power plants that would likely be built in the absence of the project activity, project participants are encouraged to submit alternative proposals for consideration by the CDM Executive Board.

[46] If 20% falls on part capacity of a unit, that unit is fully included in the calculation

The set of power units that comprises the larger annual generation should be used.

A power unit is considered to have been built at the date when it started to supply electricity to the grid.

Power plant registered as CDM project activities should be excluded from the sample group m. However, if group of power units, not registered as CDM project activity, identified for estimating the build margin emission factor includes power unit(s) that is (are) built more than 10 years ago then:

(i) Exclude power unit(s) that is (are) built more than 10 years ago from the group; and
(ii) Include grid connected power projects registered as CDM project activities, which are dispatched by dispatching authority to the electricity system.

Capacity additions from retrofits of power plants should not be included in the calculation of the build margin emission factor.

In terms of vintage of data, one of the following two options can be chosen:

Option 1: For the first crediting period, calculate the build margin emission factor *ex ante* based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

Option 2: For the first crediting period, the build margin emission factor shall be updated annually, ex-post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated *ex ante*, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

For the proposed project, option 1 is chosen to calculate Build Margin emission factor.

The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units *m* during the most recent year *y* for which power generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

(5)

Where:

$EF_{grid,BM,y}$ =Build margin CO_2 emission factor in year y (tCO_2/MWh)

$EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

$EF_{EL,m,y}$ = CO_2 emission factor of power unit m in year y (tCO_2/MWh)

m = Power units included in the build margin

y = Most recent historical year for which power generation data is available

No matter which options for calculating BM factor mentioned in step 5 was adopted for the proposed project; the same issue on data availability must be addressed. Currently, it is very difficult to get the capacity margin data of power plants in China, since these data as well as net quantity of electricity generated and delivered to the grid and fuel consumption data in power unit m are regarded as commercial secrets or only for internal usage. Then the following deviation was adopted to calculate the Build Margin emission factor.

1. The breakdown data by power plants are not while the aggregate data by different types of fuels are available. Considering this situation, the m sample group will consist of capacity addition by power sources with same fuel instead of by power plants. For the proposed project the m sample group will consist of fossil fuel fired capacity addition, hydropower capacity addition and other capacity addition;

2. To be conservative, zero emission factors were selected for hydropower capacity and other capacity. Moreover, since specific data on coal fired capacity, oil fired capacity, and gas fired capacity could not be separated from current statistical data on fossil fuel fired capacity, the following approach was adopted for calculating the emission factor of fossil fuel fired capacity addition:

(1) With the energy balance sheet in China Energy Statistical Yearbook for the most recent year, calculating the respective percentages of CO_2 emissions from coal fired power generation, oil fired power generation, and gas fired power generation against total CO_2 emissions from fossil fuel fired power generation:

$$\lambda_{Coal,y} = \frac{\sum_{i \in COAL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,j,y}} \quad (6)$$

$$\lambda_{Oil,y} = \frac{\sum_{i \in OIL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,j,y}} \quad (7)$$

$$\lambda_{Gas,y} = \frac{\sum_{i \in GAS,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO_2,i,j,y}} \qquad (8)$$

Where:

$F_{i,j,y}$ = The amount of fuel i (in a mass or volume unit) consumed by province j;

$NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i (GJ/ mass or volume unit);

$EF_{CO_2,i,j,y}$ = CO_2 emission factor of fossil fuel type i in year y (tCO_2/GJ)

$COAL$, OIl & GAS = The aggregation of various kinds of coal, oil, and gas as fossil fuels.

2) Calculating the corresponding emission factor for fossil fuel fired power generation:

$$EF_{Thermal,y} = \lambda_{Coal,y} \times EF_{Coal,Adv,y} + \lambda_{Oil,y} \times EF_{Oil,Adv,y} + \lambda_{Gas,y} \times EF_{Gas,Adv,y} \qquad (9)$$

Where:

$EF_{Coal,Adv,y}$, $EF_{Oil,Adv,y}$ & $EF_{Gas,Adv,y}$ are the emission factors for the best commercially available technology of coal fired power generation, oil fired power generation, and gas fired power generation, respectively (See Annex 3 for detailed calculation).

3. Using the share of different type of capacity in total capacity addition as weight, the weighted average of emission factors of different type capacity is calculated as the Build Margin emission factor $EF_{grid,BM,y}$ of Northeast China Power Grid (See Annex 3 for detailed calculation):

$$EF_{grid,BM,y} = \frac{CAP_{Thermal,y}}{CAP_{Total,y}} \times EF_{Thermal,y} \qquad (10)$$

Where:

CAP_{Total} = The total capacity addition

$CAP_{Thermal}$ =The fossil fuel fired capacity addition

Following the four steps above, the build margin emission factor $EF_{grid,BM,y}$ of the Northeast China Power Grid is calculated to be **0.7242tCO_2/MWh.** The detailed calculations and data are listed in the annex 3 (The build margin emission factor BM is same as that provided by Chinese DNA, the website is (http://cdm.ccchina.gov.cn/WebSite/CDM/UpFile/File2333.pdf).

Step 6: Calculate the combined margin emissions factor

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$
(11)

Where:
$EF_{grid,BM,y}$ =Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EF_{grid,OM,y}$ =Operating margin CO_2 emission factor in year y (tCO_2/MWh)
w_{OM} =Weighting of operating margin emissions factor (%)
w_{BM} =Weighting of build margin emissions factor (%)

Wind project activities: w_{OM} = 0.75 and w_{BM} = 0.25 (owing to their intermittent and non dispatchable nature) for the first crediting period and for subsequent crediting periods.

The default weights are adopted for the proposed project, the baseline emission factor is:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} = \mathbf{1.0280\ tCO_2/MWh}$$

Project emissions

For the proposed wind power project activities, there are no emissions from fossil fuel consumption, operation of geothermal power plants or water reservoirs of hydro power plants.

$$PE_y = PE_{EF,y} + PE_{GP,y} + PE_{HP,y} = 0$$
(12)

Where:
PE_y = Project emissions in year y ($t\ CO_2e/yr$).
$PE_{EF,y}$ = Project emissions from fossil fuel consumption in year y ($t\ CO_2e/yr$).
$PE_{GP,y}$ = Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y ($t\ CO_2e/yr$).
$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y ($t\ CO_2e/yr$).

Leakage

For the proposed wind power project activities, no leakage emissions are considered. The main emissions potentially giving rise to leakage in the context of electric sector projects are emissions arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, and transportation). These emissions sources are neglected.

Emission reductions

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \quad (13)$$

Where:
ER_y = Emission reductions in year y (t CO_2e/yr).
BE_y = Baseline emissions in year y (t CO_2e/yr).
PE_y = Project emissions in year y (t CO_2/yr).

BE_y is calculated using the following formula:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (14)$$

Where:
$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr).
$EF_{grid,CM,y}$ = The baseline emission factor.

$EG_{PJ,y}$ is calculated using the following formula:

$$EG_{PJ,y} = EG_{facility,y} \quad (15)$$

Where:
$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y,

Then, the annual emission reduction of the proposed project is:

$$ER_y = BE_y - PE_y = BE_y = EG_{facility,y} \times EF_{grid,CM,y} = \left(EG_{\mathrm{exported},y} - EG_{imported,y}\right) \times EF_{grid,CM,y} \quad (16)$$

B.6.2. Data and parameters that are available at validation:

Data / Parameter:	*Power generation*
Data unit:	*MWh*
Description:	*The total electricity generation and the electricity generated by those low-cost/musts run power plants of NECPG on 2003, 2004, 2005, 2006 and 2007.*
Source of data used:	*China Electric Power Yearbook 2004, 2005, 2006, 2007 and 2008 Edition.*
Value applied:	*Detailed in China Electric Power Yearbook 2004-2008 Edition.*
Justification of the choice of data or description of measurement methods and procedures actually applied :	*Official data.*
Any comment:	-

Data / Parameter:	EG_y
Data unit:	*MWh*
Description:	*The net electricity generated and delivered to NECPG on 2005, 2006 and 2007, not including those generated by low-cost/must run power plants/units.*
Source of data used:	*China Electric Power Yearbook 2006-2008 Edition.*
Value applied:	*Detailed in Annex 3.*
Justification of the choice of data or description of measurement methods and procedures actually applied :	*Official data.*
Any comment:	-

Data / Parameter:	*Installed Capacity*
Data unit:	*MW*
Description:	*The installed capacity by different sources of NECPG in year 2004, 2005 and 2007.*
Source of data used:	*China Electric Power Yearbook 2005, 2006 and 2008 Edition.*
Value applied:	*Detailed in Annex 3.*
Justification of the choice of data or	*Official data.*

description of measurement methods and procedures actually applied :	
Any comment:	

Data / Parameter:	$F_{i,j,y}$
Data unit:	*10^4t or 10^8m^3*
Description:	*Different fuel consumptions for power generation in NECPG in year 2005, 2006 and 2007.*
Source of data used:	*China Energy Statistical Yearbook 2006, 2007 and 2008 Edition.*
Value applied:	*Detailed in Annex 3.*
Justification of the choice of data or description of measurement methods and procedures actually applied :	*Official data.*
Any comment:	-

Data / Parameter:	NCV_i
Data unit:	*GJ/t or GJ/10^3m^3*
Description:	*Average low calorific values of fuels for electricity generation.*
Source of data used:	*China Energy Statistical Yearbook 2008 Edition, P283.*
Value applied:	*Detailed in Annex 3.*
Justification of the choice of data or description of measurement methods and procedures actually applied :	*China-specific values are adopted.*
Any comment:	*Official data.*

Data / Parameter:	$EF_{CO2,i}$
Data unit:	*tC/TJ*
Description:	*Emission factors of fuels for electricity generation.*
Source of data used:	*Table 1.3 and 1.4, Page 1.21-1.24, "2006 IPCC Guidelines for National Greenhouse Gas Inventories" Volume 2 Energy.*
Value applied:	*Detailed in Annex 3.*
Justification of the choice of data or	*IPCC world-wide default values are adopted. Data issued by IPCC.*

description of measurement methods and procedures actually applied :	
Any comment:	
Data / Parameter:	$EF_{Coal,Adv}$, $EF_{Oil,Adv}$ and $EF_{Gas,Adv}$
Data unit:	*tCO_2e/MWh*
Description:	*The efficiency level of the best coal-based, oil-based and gas-based power generation technology commercially available in China.*
Source of data used:	*Notification on Determining Baseline Emission Factors of China Power Grid*
Value applied:	*Detailed in Annex 3.*
Justification of the choice of data or description of measurement methods and procedures actually applied :	*Official data.*
Any comment:	

B.6.3. Ex-ante calculation of emission reductions:

According to the calculation results in B6.1, the emission reductions of the proposed project are calculated as follows:

1.1.1.1.1 Baseline emissions

With reference to the Notification on Determining Baseline Emission Factors of China Power Grid issued by Chinese DNA on July 2nd, 2009, the OM emission factor ($EF_{OM,y}$) of NECPG is 1.1293tCO_2e/MWh, and the build margin emission factor ($EF_{BM,y}$) of NECPG is 0.7242tCO_2e/MWh. The detailed calculations and data are listed in Annex 3.

The baseline emissions factor (EF_y) of NECPG is calculated with formula (9) in part B.6.1 as follow:

$$EF_y = w_{OM} \cdot EF_{OM,y} + w_{BM} \cdot EF_{BM,y} = 1.028 tCO_2e/MWh$$

Annual electricity output to NECPG of the Project is 120,069MWh. Based on the emission factors in the previous step, the baseline emission of the Project can be calculated with formula (1) in part B.6.1 as follow:

$$BE_y = 123{,}430 tCO_2e$$

Project emissions

According to the baseline methodology ACM0002, the GHG emission of the proposed project within the project boundary is zero

$$PE_y = 0$$

Emission Reductions

The emission reduction (ER_y) by the project activity during a given year y is the difference between baseline emissions (BE_y), project emissions (PE_y), as follows:

$$ER_y = BE_y - PE_y = 123{,}430 - 0 = 123{,}430 tCO_2e$$

B.6.4. Summary of the ex-ante estimation of emission reductions:

Year	Estimation of project activity emissions (tonnes of CO_2e)	Estimation of baseline emissions (tonnes of CO_2e)	Estimation of Leakage (tonnes of CO_2e)	Estimation of overall emission reductions (tonnes of CO_2e)
01/03/2012-31/12/2012	0	102,859	0	102,859
2013	0	123,430	0	123,430
2014	0	123,430	0	123,430
2015	0	123,430	0	123,430
2016	0	123,430	0	123,430
2017	0	123,430	0	123,430
2018	0	123,430	0	123,430
01/01/2019-28/02/2019	0	20,571	0	20,571
Total (tonnes of CO_2e)	0	864,010	0	864,010

B.7. Application of the monitoring methodology and description of the monitoring plan:

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B.7.1. Data and parameters monitored:

Data / Parameter:	$EG_{facility,y}$
Data unit:	MWh/yr
Description:	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data:	Measured electricity meters installed in the substation
Measurement procedures (if any):	$EG_{facility,y}$ can be calculated by $EG_{export,y}$ subtracting $EG_{import,y}$.
Monitoring frequency:	The reading of electricity meters will be continuously measured and monthly recorded.
QA/QC procedures:	Bidirectional electricity meters are used for monitoring the

	quantity of power exported and imported. Net electricity generation supplied can be obtained by the export subtract the import. The meters' accuracy will not be less than 0.5%, and will be calibrated yearly according to national standards and rules. Double checked by receipt of sales.
Any comment:	See also section B.7.2 for more details.

Data / Parameter:	$EG_{export, y}$
Data unit:	MWh
Description:	Electricity exported to the grid by the project
Source of data:	Electricity meter reading on project activity site
Measurement procedures (if any):	The designated person of the project will read and aggregate the number of the meter. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Monitoring frequency:	The reading of electricity meters will be continuously measured and monthly recorded.
QA/QC procedures to be applied:	The meters' accuracy will not be less than 0.5%, and will be calibrated yearly according to national standards and rules. The data will be directly used to calculate emission reductions. The sales receipts are used to ensure consistency.
Any comment:	See also section B.7.2 for more details.

Data / Parameter:	$EG_{import, y}$
Data unit:	MWh
Description:	Electricity utilized by the project
Source of data:	Electricity meter reading on project activity site
Measurement procedures (if any):	The designated person of the project will read and aggregate the number of the meter. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Monitoring frequency:	The reading of electricity meters will be continuously measured and monthly recorded.
QA/QC procedures:	The meters' accuracy will not be less than 0.5%, and will be calibrated yearly according to national standards and. The data will be directly used to calculate emission reductions. The sales receipts are used to ensure consistency.
Any comment:	See also section B.7.2 for more details.

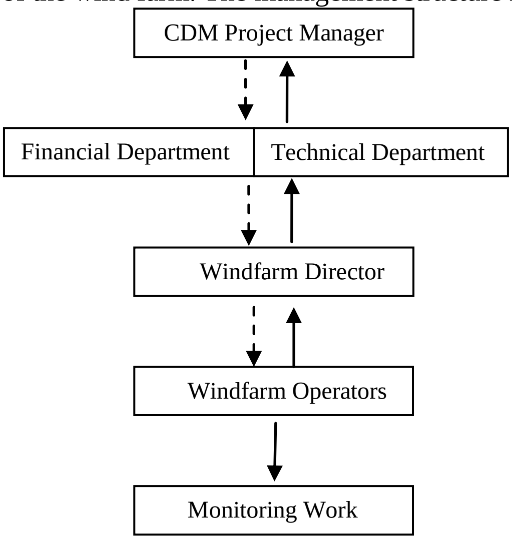
B.7.2. Description of the monitoring plan:

1. Introduction

The project adopts the approved consolidated baseline and monitoring methodology ACM0002 "Consolidated baseline methodology for grid-connected electricity generation from renewable sources (Version 12.1.0)" to determine the emission reductions from the wind farm. This plan describes in more detail the process.

2. Responsibility

Hegang Longyuan Wind Power Co., Ltd.is overall responsibility for monitoring and carries out the monitoring following this monitoring plan .The CDM manager of Hegang Longyuan Wind Power Co., Ltd. is responsible for the monitoring and reporting of the wind farm. The management structure is illustrated as follows:



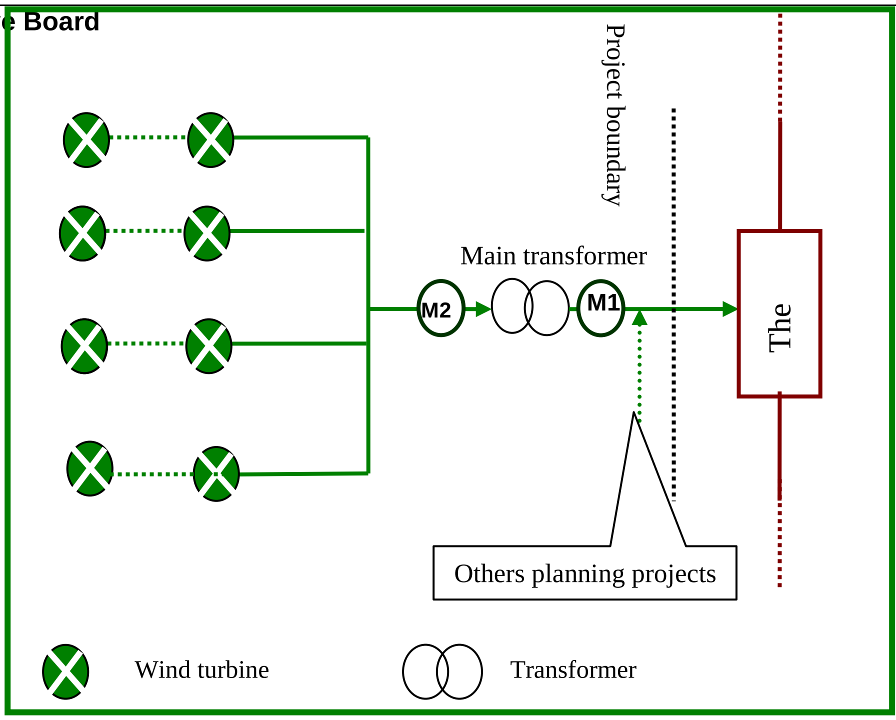
Windfarm Director will collect the information and data required by the Monitoring Plan. The collected information will be documented and sent to the CDM manager and responsible staffs of Hegang Longyuan Wind Power Co., Ltd d monthly. The CDM manager will in charge of the implementation of the Monitoring Plan and the confirmations on monitoring, calculation data and report to the General Manager of the company. Managers of the proposed project must maintain credible, transparent, and adequate data estimation, measurement, collection, and tracking systems to maintain the information required for an audit of an emission reduction project.

3. Training

Hegang Longyuan Wind Power Co., Ltd. will assign and train the dedicated people carrying out the monitoring work. The monitoring personnel training by the contacted CDM consult company will be completed before the registration, further training work will be completed with the preliminary verification.

4. Meters system

For the proposed project, a meter (M1, accuracy degree is not less than 0.5%, bidirectional) with a backup meter (M2, accuracy degree is not less than 0.5%, bidirectional) will be installed to monitor directly $EG_{import,y}$ and $EG_{export,y}$ by the proposed project (shown in the figure below).



5. Calibration

The metering equipments related with the proposed project are calibrated and tested yearly by a qualified third party appointed by the Northeast China Power Grid for accuracy according to the requirement from Technical administrative code of electric energy metering. The meters shall be jointly inspected and sealed on behalf of the parties concerned and shall not be interfered with by either party except in the presence of the other party or its accredited representatives.

6. Monitoring data

According to the illustration figure above, the net electricity supplied to the grid by the proposed project is calculated by:

$\mathbf{EG}_{facility,y} = EG_{export,y} - EG_{import,y}$

7. Quality assurance and Quality control

The quality assurance and quality control procedures for the emergency about recording, maintaining and archiving data shall be improved as part of this CDM project activity. This is an on-going process that will be ensured through the CDM in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002.

Hegang Longyuan Wind Power Co., Ltd specially issued *the regulations of monitoring and management for CDM project (*hereafter refer as *CDM Manual)* to monitor and verify the emission reductions from the proposed project. Moreover, the management procedures and regulations like operation, meter recordings, maintenance and emergency management are established in the *CDM Manual*, which guides the technicians to operate and guarantee the quality assurance and quality control procedures for recording, maintaining and archiving data in detail. This is an on-going process that will be ensured through the CDM in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002.

Several processes required to meet the requirements for emissions reduction monitoring include:

- The project owner reads the meters system monthly and supplies the recordings to the local Electric Power Company. The local Electric Power Company checks the recordings supplied by the project owner.
- If either party finds any reading of the Meters more than the allowable error or the monitoring equipment functions improperly, it should inform the other party immediately. The project owner and the local Power Grid Company should retain a qualified measure institute together to check the meters or equipment, solve the problems and get everything into normal condition; In the mean time, both sides shall jointly prepare an acceptable estimation method according to *Regulations and Rules of Power Supply*[47] issued by State Electricity Regulatory Commission of China to calculate the electricity exported to the Grid, and provide evidence to DOE for the verification to show the estimation is reasonable and conservative.

For the handling of disputes between the proposed project owner and the Grid, measures will be adopted according to relevant articles of *Interim Measures for Settlement of Electricity Charges between the Power Generating Enterprises and the Grid Enterprises*[48] issued by State Electricity Regulatory Commission of China. Moreover, if meters are out of order, the electricity generation in the period will be not considered.

8. Data Management System

Overall responsibility for monitoring greenhouse gas emissions reductions will rest with the CDM monitoring staff of the proposed project. The CDM manual sets out the procedures for tracking information from the primary source to data calculations in paper format. Moreover, the credibility and reliability of those data and information must be confirmed. Physical documentation such as paper-based maps, diagrams and environmental assessment will be collated in a central place, together with this monitoring plan. In order to facilitate auditor's reference, monitoring results will be indexed. All paper-based information will be stored by Hegang Longyuan Wind Power Co., Ltd and kept at least one copy, and all data including calibration records is kept until 2 years after the end of the total crediting period of the CDM project.

The following table below outlines the key documents relevant to monitoring and verification of the emission reductions from the proposed project.

Table List of the key documents relevant to monitoring and verification

I.D.No.	Document Title	Main Content	Source
F-1	PDD, including the electronic spreadsheets and supporting documentation(assumptions, estimations, measurement, etc)	Calculation procedure of emission reduction and monitoring items	Project owner or UNFCC website
F-2	Report on monitoring and checking of electricity supplied to the grid	Record based on monthly meter reading and electricity sale receipts	Project owner
F-3	Report on maintenance and calibration of metering equipment	Reasons for maintenance and calibration and the precision after	Prroject owner

47 http://www.serc.gov.cn/flfg/bmgz/200802/t20080220_5931.htm

48 http://www.chinapower.com.cn/article/1130/art1130580.asp

		maintenance and calibration	
F-4	Report on the qualifications of the operators	Technical post ,working experience etc.	Project owner
F-5	the project management record (including date collection and management system)	Comprehensively and truly reflect the management and the operation of the proposed project	Project owner

9. Verification and monitoring results

The verification of the monitoring results of the proposed project is a mandatory process required for all CDM projects. The main objective of the verification is to independently verify that the project has achieved the emission reductions as reported and projected in the PDD. It is expected that the verification will be done annually.

B.8. Date of completion of the application of the baseline study and monitoring methodology and the name of the responsible person(s)/entity(ies)

Date of completion of baseline study: 01/08/2011

Names of person/entity determining the baseline are listed as follows:

(Not the project participants listed in Annex 1)

Li Gang

Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd

Address: Floor 7, Part C, International Investment Building, No.6-9, Fuchengmen North Road, Xicheng District, and Beijing, China. 100034

Tel: +86-10-66091326

Fax: +86-10-66091396

Email: Ligang@clypg.com.cn

SECTION C. Duration of the project activity / Crediting period

C.1. Duration of the project activity:

C.1.1. Starting date of the project activity:

12/05/2009（Real action- the earliest date for the Turbine Purchasing contract of equipment.）

The Turbine Purchasing contract was signed on May 12th, 2009 and the transformer purchasing contract for the project was signed on May 14th, 2009. And the start work order was signed on Jun. 4th, 2009. So, the PP chooses the date of signature for earliest date as the starting date of CDM project activity conservatively.

C.1.2. Expected operational lifetime of the project activity:

20 years

C.2. Choice of the crediting period and related information:

C.2.1. Renewable crediting period

C.2.1.1. Starting date of the first crediting period:

01/03/2012

C.2.1.2. Length of the first crediting period:

7 years and 0 month.

C.2.2. Fixed crediting period:

C.2.2.1. Starting date:

Not applicable.

C.2.2.2. Length:

Not applicable.

SECTION D. Environmental impacts

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D.1.Documentation on the analysis of the environmental impacts, including transboundary impacts:

In accordance with relevant environmental law and regulations, the Environmental Assessment Report Table of the project has been approved by the Environmental Protection Administration of Heilongjiang Province, referred as “Heihuanjianshen [2008] No.164”. A summary of the report is illustrated as follows:

- **Impacts on Water**

The wind-farm does not consume any water, nor does it generate any wastewater in the operation phase. The possible negative impacts are the household wastewater. Under normal conditions with highly automated monitoring and control system, the household wastewater will be first treated in a septic tank, and then be disinfected to discharge for circumjacent virescence.

- **Impacts Solid Waste**

Solid waste produced by builders and staff, and the waste earth from digging of the foundation in the construction phase. The amount of household solid waste will be very little, which will not have impact on the environment. Besides, the solid waste will be collected and moved to the landfill site of the nearest city. The waste earth from the digging should be firstly used for refilling. The rest of the waste earth should be placed in the low area of the site and replanted with grass.

- **Impacts on Noise Environment**

The noise of the project in construction phase is from vehicles and machines on-site. According to the monitoring data from the construction site, the noise is at a level between 85-100dB.

Moreover, the magnitude of the impacts during construction phase only exists for a temporary period and will expire with the end of the construction phase. Moreover, operational noise from the rotating blades is expected to be minimal due to the higher background noise caused by strong winds. The closest residential area to the site of the Project is over 1km away. Therefore, the noise of the project will not have impact on nearby residents.

Impacts on Air Environment

Wind Power plants are known to contribute to zero atmospheric pollution as no fuel combustion is involved during any stage of the operation. However, the sources of air pollution are mainly due to the construction activities including the transportation of construction material, road construction and Improvement and cadre construction etc. The impacts on air environment are temporarily that the impact will be ended when the construction is completed. It is suggested that several measures shall be taken into account, such as the construction under strong wind weather is prohibited, reducing the area of construction as much as possible, spraying water when undertaking construction, and reducing the speed of vehicles in the field. Hence, air pollution caused by the project is not significant to the surrounding environment.

- **Impacts on Ecosystem Environment**

A serious potential concern for wind farms is their impact on vegetation, animals and migrating birds. The land on which the project activity takes place is barren and unfertile. Prior to the project activity the land had no beneficial use. The vegetation in the project area was substituted by grassland for livestock use and land for cultivation. So the minor quantity of solid / liquid discharge, likely to be generated during the construction phase has no noticeable impact on soil use and the project proponent has made arrangements to dispose them in an environmentally acceptable manner. Moreover, there are no migratory birds / endangered species in the region of project activity. Therefore, the activities to be carried out will not generate any negative impact on the ecological environment.

- **Ecological impacts**

The impacts on the vegetation and the use of land are mainly from activities of construction. The plants destroyed due to construction activities are mostly secondary shrubbery and weed, and there have no impacts on valuable and rare plants. With the implementation of water and solid conservation plan, the impacted vegetation can be recovered. Therefore, the Project has little impacts on the regional ecosystem and the vegetation.

In conclusion, environmental impacts arising from the Project are considered insignificant.

D.2. If environmental impacts are considered significant by the project participants or the host Party, please provide conclusions and all references to support documentation of an environmental impact assessment undertaken in accordance with the procedures as required by the host Party:

The Project will not have significant impacts on local environment in general. Meanwhile, the EIA of the Project has been approved by the local environmental protection administration. In conclusion, environmental impacts arising from the Project are considered insignificant.

SECTION E. Stakeholders' comments

E.1. Brief description how comments by local stakeholders have been invited and compiled:

In May 2009, staff from the project owner carried out a survey of local residents possibly be impacted in the area where the Project is sited to collect public comments and attitudes towards the Project before the construction starting. The comments on the project activity by the local stakeholders have been invited and complied in two ways:

1) Symposium

On May 6th, 2009, under the support of local government, the project owner successfully held a stakeholder symposium in Luobei County. Totally 10 stakeholder representatives participated the symposium, respectively from the Development and Reform Bureau of Luobei County, several other department of local government and some villagers around plant. The project participants informed them about the project, asked for their comments on the project concerning socio-economic and environmental aspects, namely as follows,

1) Impact on the economic aspect, including the local economy, income and employment, etc;
2) Impact on the environmental aspect, including the ecological environment, air, noise and the impact of soil erosion, etc;
3) Impact on sustainable development;
4) Suggestions and recommendations on the proposed project;
5) Attitude to the implementation of the proposed project, whether or not to support.

Summary of the comments received in the symposium:

- All the stakeholder representatives believe that the proposed project will have positive impacts on economy, environment and social in local region.

- The local government highly supports the proposed project, and expects the increase of local financial incoming, which can provide more public health services and rebuild schools, etc.
- All stakeholders think that the proposed project will improve the life quality of local residents, such as, providing more employment opportunities for local residents, increasing incomes of the local residents, increasing revenues from tourism with the wind farm as a beautiful landscape, accelerate the development of local service industry, and tourism.
- All stakeholders think that the proposed project activity has no significant negative influence on local environment.
- All the stakeholder representatives support and welcome the proposed project.

2) Questionnaire

In order to finish this survey, project owner designed the style of Questionnaire and visited to Damahe Forest Farm, Luobei County to do door to door Interviews before the **Symposium**. The investigators went into the house of local residents to finish the work face to face. The local residents mainly included some villagers inhabiting in the town. The survey was conducted through distributing and collecting responses to a questionnaire.

The following is the main questions of the questionnaire.

- ✧ Do you know the proposed project?
- ✧ What do you think the proposed project bring impacts to local economy development?
- ✧ What do you think the proposed project bring impacts to incensement of employment opportunities?
- ✧ What do you think the proposed project bring impacts to of local environment during the construction period?
- ✧ What do you think the proposed project bring impacts to of local environment during the operation period?
- ✧ What do you think the proposed project bring positive impacts to local residents' living?
- ✧ What do you think the proposed project bring negative impacts to local residents' living?
- ✧ Do you support the construction of the proposed project?
- ✧ What other advices do you have for the proposed project?

Questionnaires were distributed according to the principle of both representation and randomness in order to reflect the public opinions and comments in a fair and real manner. Totally 40 questionnaires returned out of 40 with 100% response rate.

Table10 the Statistic of the respondents

Item		Number	Percentage (%)
Sex	Male	35	87.5%
	Female	5	12.5%
Age	≤30	1	2.5%
	31-40	18	45%
	41-50	19	47.5%
	>50	2	5%
Education	Junior high school and below	29	72.5%
	senior high school and above	11	27.5%

E.2. Summary of the comments received:

The following is a summary of the key findings based on 40 returned questionnaires.

- ✧ 40 respondents (100%) know the proposed project
- ✧ 40 respondents (100%) think that the proposed project brings positive impacts to local economy development.

- ✧ 40 respondents (100%)think the proposed project bring positive impacts to incensement of employment opportunities
- ✧ 40 respondents (100%) think that the proposed project brings mild impacts to of local environment during the construction period?
- ✧ 40 respondents (100%) think that the proposed project brings mild impacts to of local environment during the operation period.
- ✧ 40 respondents (100%) think that bring several positive impacts to local residents' living, such as lessening of power shortage; increasing of income; improvement standard of living; improvement of local environment; increasing of employment opportunities.
- ✧ 12 respondents (30%) think that bring several negative impacts to local residents' living, such as impacts of water quality; ecological environment; construction noise; quality of the air.
- ✧ 40 respondents (100%) support the construction of the proposed project.
- ✧ 40 respondents (100%) don't have other advices for the proposed project except the lists on the questionnaire.

It shows that the local residents strongly support the proposed project, and they consider the proposed project will bring various positive impacts on their lives. However, parts of the local residents were concerned about the impacts on the local environment.

E.3. Report on how due account was taken of any comments received:

The Project owner will pay much attention to the comments and suggestions of stakeholders and will put all of the measures listed in the EIA into effect during construction and operation period, so as to achieve environmental benefits, social benefits and economic benefits.
To sum up, the local residents are very supportive on the Project. The Project owner has taken full consideration of the comments and suggestions given by stakeholders during the project implementation. The Project owner will also keep regular communication with the public regarding the construction and operation of the Project.

Annex 1

CONTACT INFORMATION ON PARTICIPANTS IN THE PROJECT ACTIVITY

Project Entity:

Organization:	Hegang Longyuan Wind Power Co., Ltd.
Street/P.O.Box:	No.6-9, Fuchengmen North Street, Xicheng District
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E-Mail:	/
URL:	/
Represented by:	Huang Qun
Title:	/
Salutation:	Mr
Last Name:	Huang
Middle Name:	/
First Name:	Qun
Department:	/
Mobile:	/
Direct FAX:	+86-10-66091317
Direct tel:	+86-10-66091396
Personal E-Mail:	carbon@clypg.com.cn

CERs Buyer:

Organization:	Total Gas & Power Ltd
Street/P.O.Box:	10 Upper Bank Street, Canary Wharf
Building:	/
City:	London
State/Region:	/
Postfix/ZIP:	E14 5BF
Country:	United Kingdom of Great Britain and Northern Ireland
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Annex 2

INFORMATION REGARDING PUBLIC FUNDING

No public funding from Annex I countries is involved in the proposed project.

Annex 3

BASELINE INFORMATION

Table A- 1 Annual thermal power electricity generation in NECPG in 2005

Province	Electricity generation (MWh)	Self usage rate (%)	Electricity delivered to the grid (MWh)
Liaoning	83697000	7.03	77813101
Jilin	35294000	6.59	32968125
Heilongjiang	58000000	7.96	53383200
Total			**164164426**

Data source：China Electric Power Yearbook 2006.

Table A- 2 Annual thermal power electricity generation in NECPG in 2006

Province	Electricity generation (MWh)	Self usage rate (%)	Electricity delivered to the grid (MWh)
Liaoning	96282000	6.62	89908132
Jilin	38576000	6.78	35960547
Heilongjiang	62964000	7.85	58021326
Total			**183890005**

Data source：China Electric Power Yearbook 2007.

Table A- 3 Annual thermal power electricity generation in NECPG in 2007

Province	Electricity generation (MWh)	Self usage rate (%)	Electricity delivered to the grid (MWh)
Liaoning	106500000	7.00	99045000
Jilin	43700000	7.68	40343840
Heilongjiang	68400000	7.67	63153720
Total			**202542560**

Data source：China Electric Power Yearbook 2008.

The key parameters in OM and BM calculation include the net caloric values (NCV_s) and CO_2 emission factor per unit of energy (EF_{CO_2s}) of various types of fuels, which are shown in the table below:

Table A-4: NCV_s and EF_{CO_2s} of various types of fuels

Fuel	NCV_s	EF_{CO_2s} (tc/TJ)
Coal	20908 kJ/kg	25.80
Washed coal	26344 kJ/kg	25.80
Other Washed Coal[1]	8363 kJ/kg	25.80
Coke	20908 kJ/kg	26.60
Crude oil	28435 kJ/kg	20.00
Gasoline	41816 kJ/kg	18.90
Kerosene	43070 kJ/kg	18.90
Diesel	43070 kJ/kg	20.20
Fuel oil	42652 kJ/kg	21.10
Other petroleum products[2]	41816 kJ/kg	20.00
Other coked product	38369 kJ/kg	25.80
Natural gas	28435 kJ/kg	15.30
Coke oven gas[3]	38931 kJ/m^3	12.10
Other gas[4]	16726 kJ/m^3	12.10
LPG	5227 kJ/m^3	17.20
Refinery gas	50179 kJ/kg	15.70

Data sources:
NCV_s are from China Energy Statistical Yearbook 2008
EF_{CO_2s} are from 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, table 1-3.

[1] Other washed coal includes middlings and slimes. The NCV value of middlings is adopted here, which is conservative because the NCV value of slimes is higher than that of middlings.

[2] The NCV value of other petroleum products are not provided in China Energy Statistical Yearbooks. This Annex calculates it as 38369 kJ/kg, i.e., 1.3108 tce/t, on the basis of Energy Balance Sheets (physical quantity) and conversion factor against SCE.

[3] The NCV value here adopts the lower limit of the NCV value range, i.e., 16726-17981 kJ/m^3, for coke oven gas provided in China Energy Statistical Yearbook 2008, P 283.

[4] The NCV value here adopts the lowest NCV value among those for gas by furnace, gas by heavy oil catalytic cracking, gas by heavy oil catalytic thermal cracking, gas by pressure gasification, and water coal gas, which are provided in China Energy Statistical Yearbook 2008.

Firstly, the data using in calculations of OM of NECPG are shown in several tables:

Table A-5: The fuel consumption and total emissions of Northeast China Power Grid in 2005

Fuel types	unit	Liaoning	Jilin	Heilongjiang	Sub-Total	Carbon content	Carbon Oxid	Fuel Emission Factor	Net caloric value	CO_2 emission（tCO_2e）
						(tc/TJ)	(%)	(kgc/TJ)	MJ/t,km3	I=D*G*H/100000(quantity)
		A	B	C	D=A+B+C	E	F	G	H	I=D*G*H/10000(volume)
Coal	10^4t	4305.41	2446.13	3383.21	**10134.75**	25.8	100	87,300	20908	184,986,389
Washed coal	10^4t				**0**	25.8	100	87,300	26344	0
Other Washed Coal	10^4t	524.74	19.26	24.16	**568.16**	25.8	100	87,300	8363	4,148,079
Coke	10^4t				**0**	29.2	100	95,700	28435	0
Coke oven gas	10^8M^3	1.03	3.57	0.68	**5.28**	12.1	100	37,300	16726	329,409
Other gas	10^8M^3	12.62	8.37		**20.99**	12.1	100	37,300	5227	409,236
Crude oil	10^4t	1.16			**1.16**	20	100	71,100	41816	34,488
Gasoline	10^4t				**0**	18.9	100	67,500	43070	0
Diesel	10^4t	1.18	1.48	0.57	**3.23**	20.2	100	72,600	42652	100,018
Fuel oil	10^4t	9.32	2.46	1.55	**13.33**	21.1	100	75,500	41816	420,842
LPG	10^4t	0.12			**0.12**	17.2	100	61,600	50179	3,709
Refinery gas	10^4t	5.48		1.32	**6.8**	15.7	100	48,200	46055	150,950
Natural gas	10^8M^3		0.84	2.24	**3.08**	15.3	100	54,300	38931	651,098
Other oil product	10^4t				**0**	20	100	75,500	41816	0

Other coked product	10^4t				**0**	25.8	0	95,700	28435	0
Other energy	10^4t	16.18			**16.18**	0		0	0	0
Total										191,234,218

China Energy Statistical Yearbook 2006

Table A-6: The fuel consumption and total emissions of Northeast China Power Grid in 2006

Fuel types	unit	Liaoning	Jilin	Heilongjiang	Sub-Total	Carbon content	Carbon Oxid	Fuel Emission Factor	Net caloric value	CO_2 emission（tCO_2e）
						(tc/TJ)	(%)	(kgc/TJ)	MJ/t,km3	I=D*G*H/100000(quantity)
		A	B	C	D=A+B+C	E	F	G	H	I=D*G*H/10000(volume)
Coal	10^4t	4681.99	2738.24	3698.29	**11118.52**	25.8	100	87,300	20908	202,942,832
Washed coal	10^4t	0.03			**0.03**	25.8	100	87,300	26344	690
Other Washed Coal	10^4t	674.74	17.83	96	**788.57**	25.8	100	87,300	8363	5,757,270
Coke	10^4t	3.32			**3.32**	29.2	100	95,700	28435	90,345
Coke oven gas	10^8M^3	2.68	0.16	1.44	**4.28**	12.1	100	37,300	16726	267,021
Other gas	10^8M^3	55.26	1.43		**56.69**	12.1	100	37,300	5227	1,105,268
Crude oil	10^4t	0.49			**0.49**	20	100	71,100	41816	14,568
Gasoline	10^4t				**0**	18.9	100	67,500	43070	0
Diesel	10^4t	0.75	0.39	0.3	**1.44**	20.2	100	72,600	42652	44,590
Fuel oil	10^4t	11.73	0.45	1.44	**13.62**	21.1	100	75,500	41816	429,998
LPG	10^4t				**0**	17.2	100	61,600	50179	0
Refinery gas	10^4t	8.55		4.27	**12.82**	15.7	100	48,200	46055	284,585
Natural gas	10^8M^3		0.19	2.1	**2.29**	15.3	100	54,300	38931	484,095
Other oil product	10^4t				**0**	20	100	75,500	41816	0
Other coked	10^4t				**0**	25.8	0	95,700	28435	0

product										
Other energy	10^4t	12.16		82.77	**112.53**	0		0	0	0
Total										211,421,263

China Energy Statistical Yearbook 2007

Table A-7: The fuel consumption and total emissions of Northeast China Power Grid in 2007

Fuel types	unit	Liaoning	Jilin	Heilongjiang	Sub-Total	Carbon content	Carbon Oxid	Fuel Emission Factor	Net caloric value	CO_2 emission（tCO_2e）
						(tc/TJ)	(%)	(kgc/TJ)	MJ/t,km3	I=D*G*H/100000(quantity)
		A	B	C	D=A+B+C	E	F	G	H	I=D*G*H/10000(volume)
Coal	10^4t	4869.32	2873.45	3736.11	**11478.88**	25.8	100	87,300	20908	209,520,369
Washed coal	10^4t				**0**	25.8	100	87,300	26344	0
Other Washed Coal	10^4t	747.85	16.52	106.81	**871.18**	25.8	100	87,300	8363	6,360,397
Moulded coal	10^4t				**0**	25.8	100	87,300	20908	0
Coke	10^4t	4.99			**4.99**	26.6	100	95,700	28435	135,789
Coke oven gas	10^8M^3	5.53	1.44	1.89	**8.86**	12.1	100	37,300	16726	552,758
Other gas	10^8M^3	68.38	9.06		**77.44**	12.1	100	37,300	5227	1,509,825
Crude oil	10^4t	0.24			**0.24**	20	100	71,100	41816	7,135
Gasoline	10^4t				**0**	18.9	100	67,500	43070	0
Diesel	10^4t	0.96	0.39	0.47	**1.82**	20.2	100	72,600	42652	56,357
Fuel oil	10^4t	8.43	0.45	1.48	**10.36**	21.1	100	75,500	41816	327,076
LPG	10^4t				**0**	17.2	100	61,600	50179	0
Refinery gas	10^4t	7.33		1.99	**9.32**	15.7	100	48,200	46055	206,890
Natural gas	10^8M^3		0.02	2.03	**2.05**	15.3	100	54,300	38931	433,360
Other oil product	10^4t	0.01			**0.01**	20	100	75,500	41816	316

Other coked product	10^4t	0.46			**0.46**	25.8	0	95,700	28435	12,518
Other energy	10^4t	12.41	2.43	51.35	**66.19**	0		0	0	0
Total										219,122,791

China Energy Statistical Yearbook 2008

Based on the data of 2006, 2007 and 2008, OM factor of NECPG in weighted average of thermal generation are shown in Table A-8.

Table A-8: OM factor of Northeast China Power Grid

Years	**Thermal generation delivered to NECPG**	**The emissions from NECPG**	**OM**
	A	**B**	**C=B/A**
2005	164,164,426	191,234,218	1.16489
2006	183,890,005	211,421,263	1.14972
2007	202,542,560	219,122,791	1.12928
Average OM	550,596,991	621,778,272	**1.12928**

2. BM emission factor calculation of NECPG.

Table A-9 Emission factor of the unit applying best commercially available technology

Technology	**Electricity supply efficiency**	**EF_{co2} (kgc/TJ)**	**Oxid**	**Emission factor (tCO_2/MWh)**
	A	**B**	**C**	**D=3.6/A/1000*B*C**
Coal fired plant	38.10%	87300	1	$EF_{Coal,Adv}$ =**0.8249**
Gas fired plant	49.99%	75500	1	$EF_{Gas,Adv}$ =**0.5437**
Oil fired plant	49.99%	54300	1	$EF_{Oil,Adv}$ =**0.3910**

Table A-10 Calculation of the respective percentages of CO2 emissions from coal fired power generation, oil fired power generation, and gas fired power generation against total CO2 emissions from fossil fuel fired power generation

Fuel Types	Unit	Liaoning	Jilin	Heilongjiang	Sub-Total	Net caloric value	EF_{CO2}	Oxid	CO_2 emission（tCO_2e）
		A	B	C	D=A+B+C	H	G	F	I=D*G*H/1000 (quantity)
Coal	10^4t	4869.32	2873.45	3736.11	**11478.88**	20908	87300	1	209520369
Washed coal	10^4t				**0**	26344	87300	1	0
Other Washed Coal	10^4t	747.85	16.52	106.81	**871.18**	8363	87300	1	6360397
Moulded Coal	10^4t				**0**	20908	87300	1	0
Coke	10^4t	4.99			**4.99**	28435	95700	1	135789
Other Coking Product	10^4t	0.46			**0.46**	28435	95700	1	12518
Sub-Total									216029074
Crude Oil	10^4t	0.24			**0.24**	41816	71100	1	7135
Gasoline	10^4t				**0**	43070	67500	1	0
Diesel	10^4t	0.96	0.39	0.47	**1.82**	42652	72600	1	56357
Fuel Oil	10^4t	8.43	0.45	1.48	**10.36**	41816	75500	1	327076
Other Oil Product	10^4t	0.01			**0.01**	41816	75500	1	316
Sub-Total									390885
Natual Gas	10^7M^3		0.2	20.3	**20.5**	38931	54300	1	433360
Coke Oven Gas	10^7M^3	55.3	14.4	18.9	**88.6**	16726	37300	1	552758
Other Gas	10^7M^3	683.8	90.6		**774.4**	5227	37300	1	1509825
LPG	10^4t				**0**	50179	61600	1	0
Refinery Gas	10^4t	7.33		1.99	**9.32**	46055	48200	1	206890
Sub-Toal									2702833
Total									219122791

China Energy Statistical Yearbook 2008

With the above table and formula (4), (5), and (6), the following results are achieved:

λ_{Coal} =98.58% λ_{Co}=0.18% λ_{Gas}=1.23%

$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal,Adv} + \lambda_{Oil} \times EF_{Oil,Adv} + \lambda_{Gas} \times EF_{Gas,Adv}$=**0.8190 tCO_2/MWh**

Table A-11: Capacity addition from 2004 to 2007 in the Northeast China Power Grid

	Installed capacity in 2004 (MW)	**Installed capacity in 2005 (MW)**	**Installed capacity in 2007 (MW)**	**Addition capacity(2004-2007) (MW)**	**Addition share**
	A	**B**	**C**	**D=C-A**	**E**
Thermal Power	32178.1	33934	41380	9201.9	88.42%
Hydro Power	5849.9	5971.4	6170	320.1	3.08%
Nuclear power	0	0	0	0	0.00%
Wind Power	217.4	273.3	1103	885.6	8.51%
Total	**38245.4**	**40178**	**48653**	**10407.6**	**100%**
Share of 2006 installed capacity	78.61%	82.58%	100.00%	21.39%	

Data sources: China Electric Power Yearbook 2005-2008

$EF_{grid,BM,y} = EF_{Thermal,Adv} \times CAP_{Thernal,addition} / CAP_{Total,addition}$= 0.8190×88.42%=**0.7242$tCO_2$/MWh**

3. The combined emission factor calculation of the Northeast China Power Grid

Table A-12: Combined emission factor of Northeast China Power Grid

OM factor (tCO_2/MWh)	1.1293
BM factor (tCO_2/MWh)	0.7242
CM factor (tCO_2/MWh) **CM**=0.75×**OM**+0.25×**BM**	**1.0280**

1.2 Annex 4

MONITORING PLAN

Please refer to section B.7. No need to complement more information here.

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Clarification of clauses 1.2, 1.3, 1.5, 1.6, 1.7, 1.9, 1.10, 1.12.1, 1.12.2, 1.12.3, 1.12.4 and 1.13, to the VCS PD

According to clause 3.12.8 1) of the VCS Standard v3.2 which came into force on 1 February 2012, “for projects validated under the CDM, the cover page and sections 1.2, 1.3, 1.5, 1.6, 1.7, 1.9, 1.10, 1.12.1, 1.12.2, 1.12.3, 1.12.4 and 1.13 of the *VCS Project Description Template* shall be completed.” Below are the clarifications of the designated clauses in latest *VCS Project Description Template*.

1.2 Sectoral Scope and Project Type

The project is not a grouped project.

The Project comes under sectoral scope 1, Energy Industries renewable sources.

Project Category: Renewable electricity in grid connected applications

1.3 Project Proponent

Project Proponent	Hegang Longyuan Wind Power Co., Ltd.
Attention	Wang Zhewei
Title	Project manager
Direct Tel	0451 51919055
Email address	wangzhewei@clypg.com.cn

1.5 Project Start Date

The Project Start Date is 20/01/2010, when it started generating emission reductions.

1.6 Project Crediting Period

The Project crediting period is from 20/01/2010 to 30/01/2012, covering 2 years.

1.7 Project Scale and Estimated GHG Emission Reductions or Removals

Project	yes
Mega-project	no

Years	Estimated GHG emission reductions or removals (tCO_2e)

2010	**123,430**
2011	**123,430**
Total estimated ERs	**246,860**
Total number of crediting years	**2**
Average annual ERs	**123,430**

1.9 Project Location

The proposed project is located in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, and P.R.China. Its geographical coordinates are north latitude 47° 54′ 34′ ′ , and east longitude 130° 39′ 56′ ′ , its altitude is about 240m~430m.

1.10 Conditions Prior to Project Initiation

The project is a renewable energy generation project, which discharges no emission during operation period. Thus, the project doesn't fall into categories that creating GHG emissions primarily for the purpose of its subsequent removal or destruction.

1.12.1 Proof of Title

The project including the plant, equipment and its emissions reductions are owned by Hegang Longyuan Wind Power Co., Ltd. Relevant evidences for the ownership will be provide to the DOE, which involves business license of the project company, Approval of the project from Chinese National Development and Reform Commission (NDRC) and Letter of Approval for the project as a CDM project by Chinese NDRC.

1.12.2 Emissions Trading Programs and Other Binding Limits

The GHG emission reductions generated by the project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions.

1.12.3 Participation under Other GHG Programs

The project was registered as a CDM project on 31/01/2012 for which a renewable crediting period of 3×7 years will be used under the CDM GHG Program. Therefore, CO_2 emission reductions generated by the project during the CDM crediting period will be verified as unique CERs but not

VCUs to avoid double counting. As to the project under VCS Standard v3.2, only emission reductions achieved from 20/01/2010-30/01/2012 will be considered as VERs.

1.12.4 Other Forms of Environmental Credit

The project has not created another form of environmental credit, which will be verified by DOE.

1.13 Additional Information Relevant to the Project

Eligibility Criteria

The project is not a grouped project.

Leakage Management

According to the CDM baseline methodology ACM0002, the leakage of the proposed project is not considered for this project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

NA